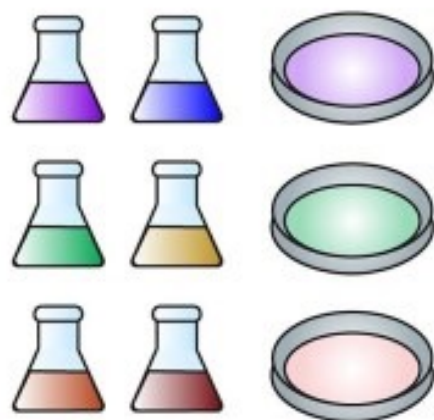
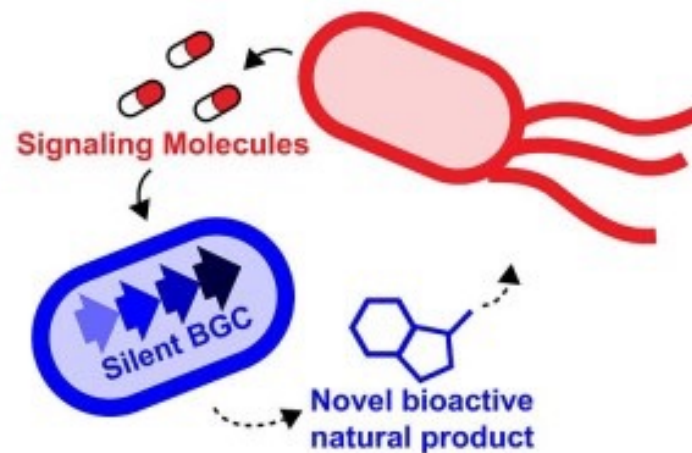
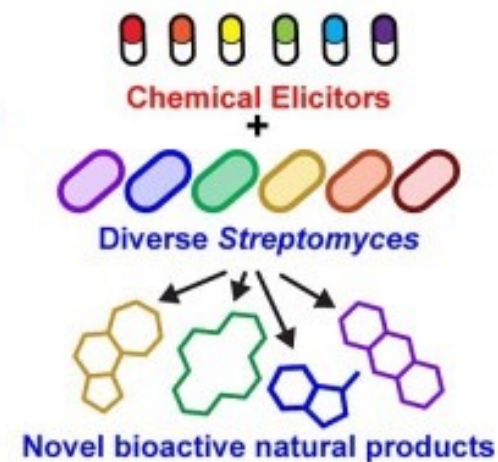
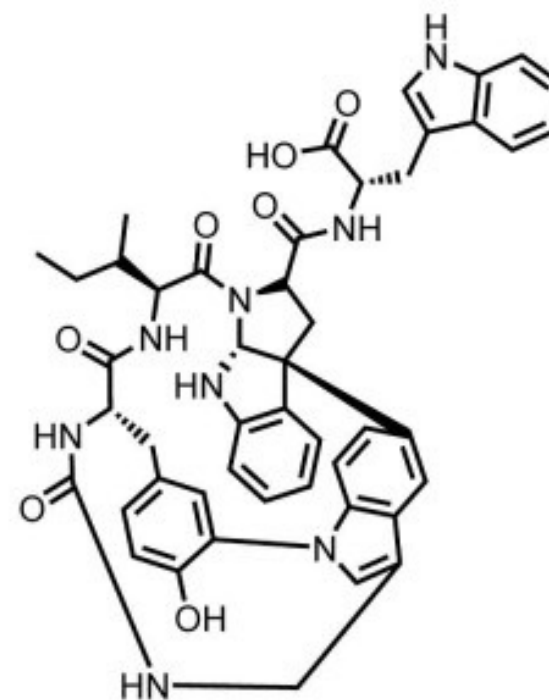
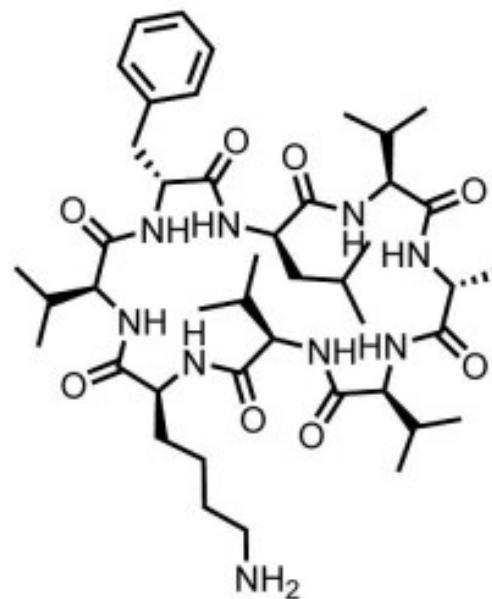
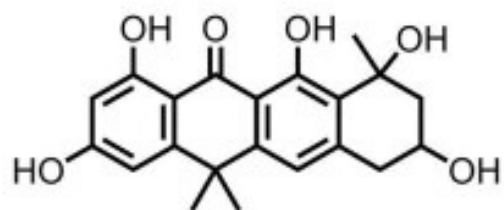


(A)

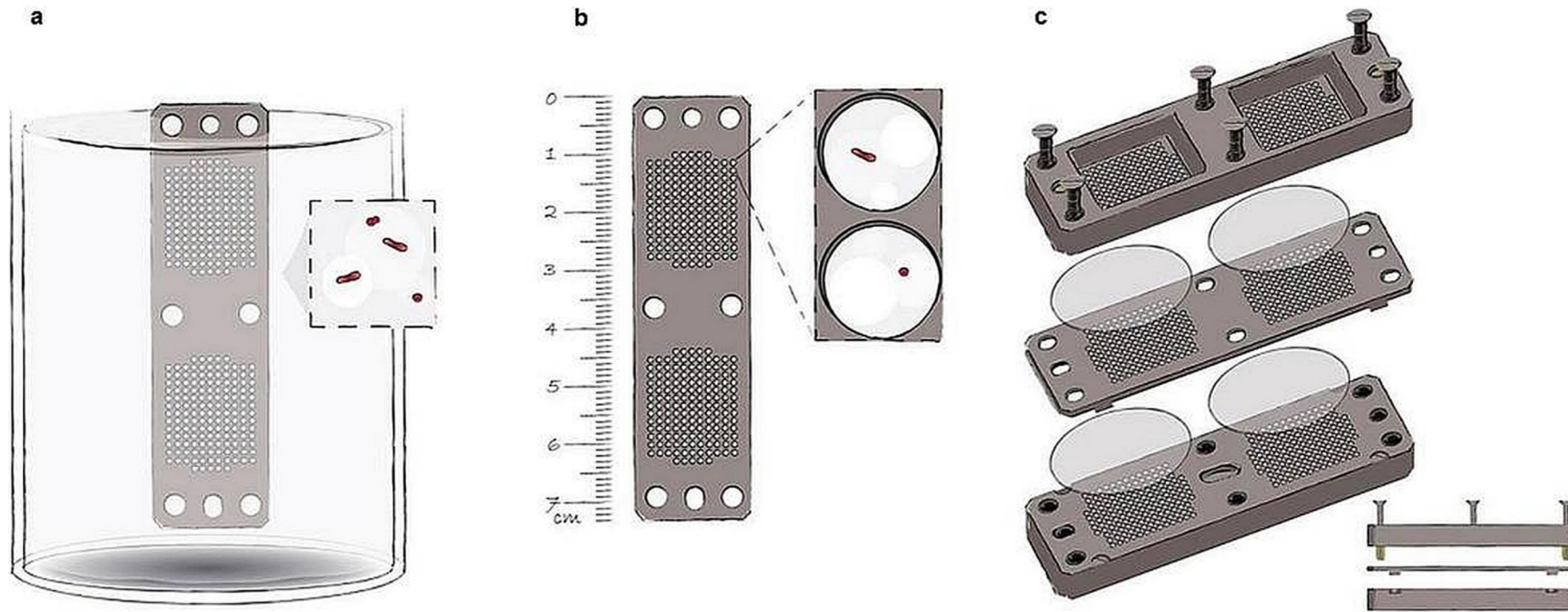
OSMAC**Co-Culture****Chemical Elicitors**

(B)



Culturing the unculturable – iCHIP technology

Uncultured bacteria make up approximately 99% of all species in external environments, and are an untapped source of new antibiotics.

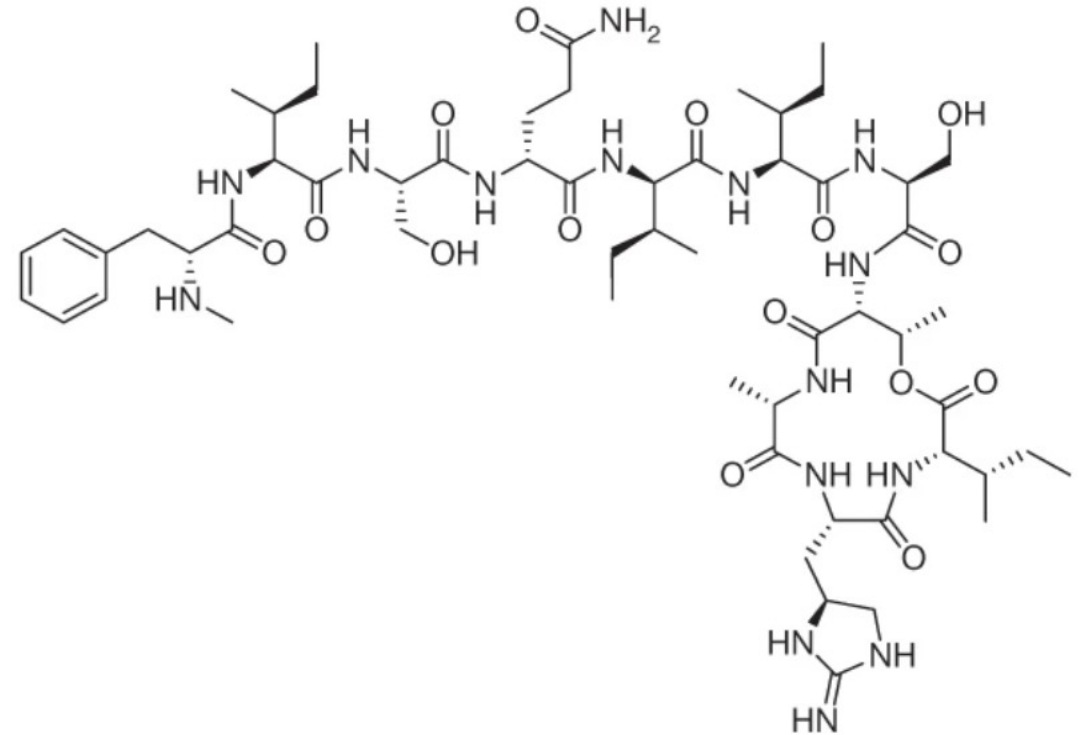


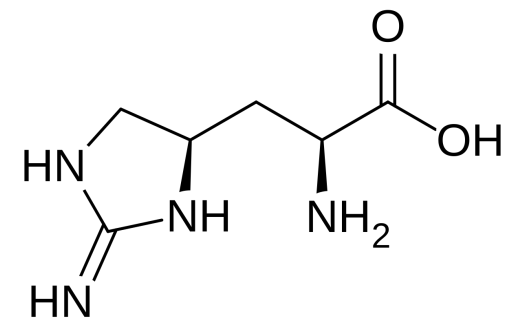
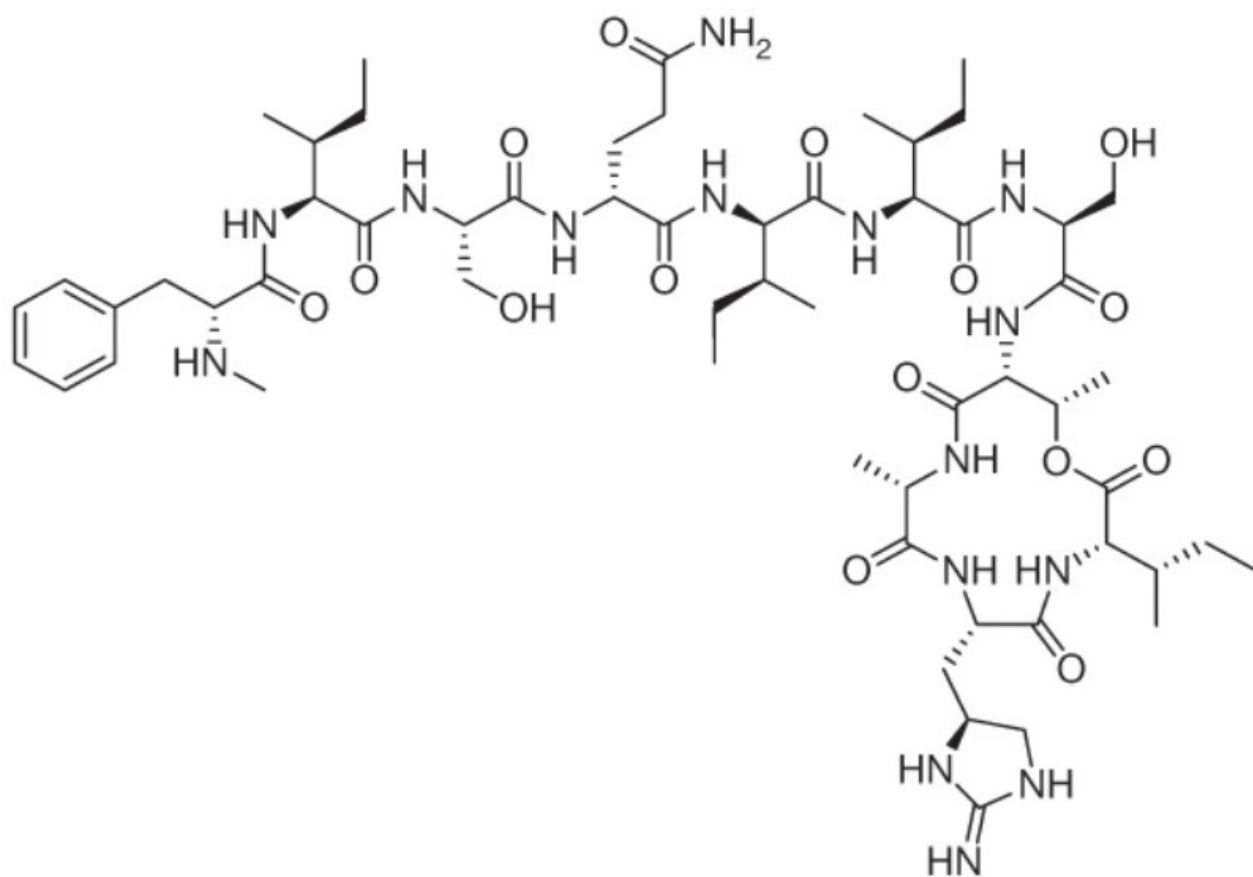
Teixobactin – first novel antibiotic discovered through iCHIP.

Extracts from 10,000 isolates obtained by growth in iChips were screened for antimicrobial activity on plates overlaid with *S. aureus*. An extract from a new species of β -proteobacteria provisionally named *Eleftheria terrae* showed good activity. The genome of *E. terrae* was sequenced.

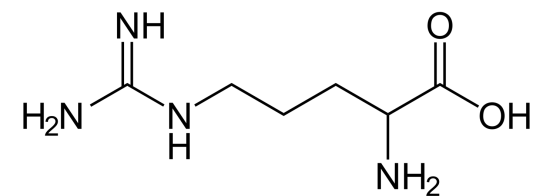
Based on 16S rDNA and *in silico* DNA/DNA hybridization, this organism was reported to belong to a new genus related to *Aquabacteria*. This group of Gram-negative organisms is not known to produce antibiotics.

A partially purified active fraction contained a compound with a molecular mass of 1,242 Da determined by mass spectrometry, which was not reported in available databases. The compound was isolated and a complete stereochemical assignment has been made based on NMR and advanced Marfey's analysis.





Enduracididine – a cyclic analog of arginine



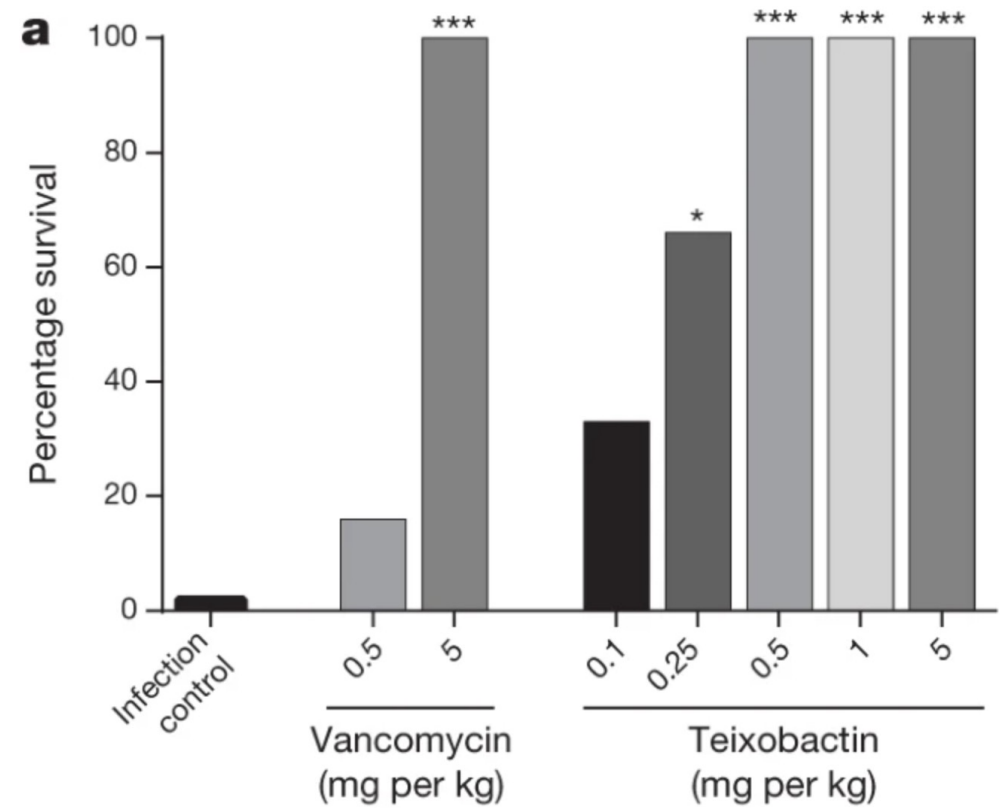
Arginine

Teixobactin had excellent activity against Gram-positive pathogens, including drug-resistant strains. Teixobactin inhibits cell wall synthesis by binding to a highly conserved motif of lipid II (precursor of peptidoglycan) and lipid III (precursor of cell wall teichoic acid).

Organism and genotype	Teixobactin MIC (µg ml ⁻¹)
<i>S. aureus</i> (MSSA)	0.25
<i>S. aureus</i> + 10% serum	0.25
<i>S. aureus</i> (MRSA)	0.25
<i>Enterococcus faecalis</i> (VRE)	0.5
<i>Enterococcus faecium</i> (VRE)	0.5
<i>Streptococcus pneumoniae</i> (penicillin ^R)	≤ 0.03
<i>Streptococcus pyogenes</i>	0.06
<i>Streptococcus agalactiae</i>	0.12
Viridans group streptococci	0.12
<i>B. anthracis</i>	≤ 0.06
<i>Clostridium difficile</i>	0.005
<i>Propionibacterium acnes</i>	0.08
<i>M. tuberculosis</i> H37Rv	0.125
<i>Haemophilus influenzae</i>	4
<i>Moraxella catarrhalis</i>	2
<i>Escherichia coli</i>	25
<i>Escherichia coli</i> (asmB1)	2.5
<i>Pseudomonas aeruginosa</i>	>32
<i>Klebsiella pneumoniae</i>	>32

Teixobactin inhibited peptidoglycan biosynthesis reactions *in vitro* in a dose-dependent manner with either lipid I, lipid II or undecaprenyl-pyrophosphate . Teixobactin was active against vancomycin-resistant enterococci that have modified lipid II (lipid II-D-Ala-D-Lac or lipid II-D-Ala-D-Ser instead of lipid II-D-Ala-D-Ala).

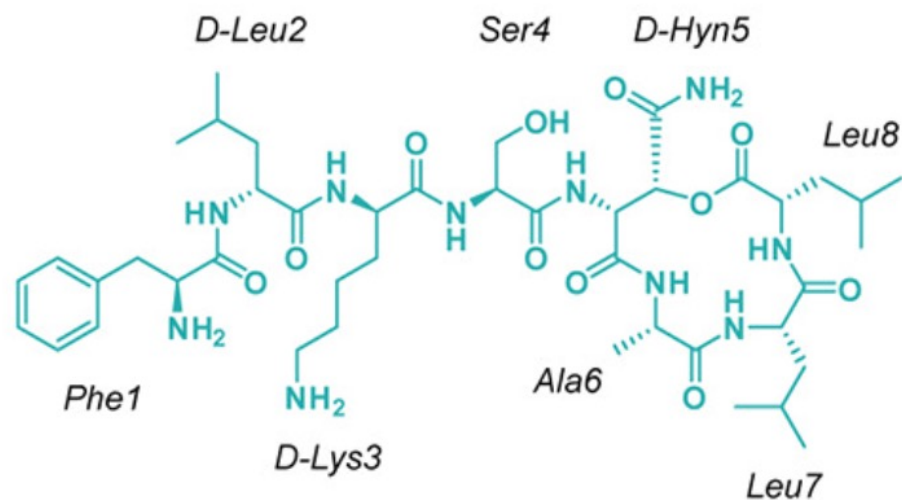
Mice were infected intraperitoneally with methicillin-resistant *S. aureus* (MRSA) at a dose that leads to 90% of death. One hour post-infection, teixobactin was introduced i.v. at different doses.



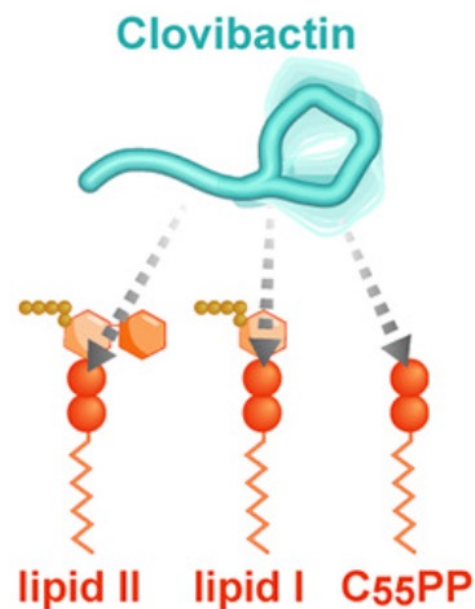
An antibiotic from an uncultured bacterium binds to an immutable target

In 2023, Clovibactin was discovered by NovoBiotic Pharmaceuticals and microbiologist Kim Lewis, PhD, professor at Northeastern University.

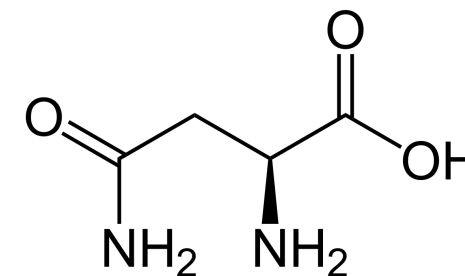
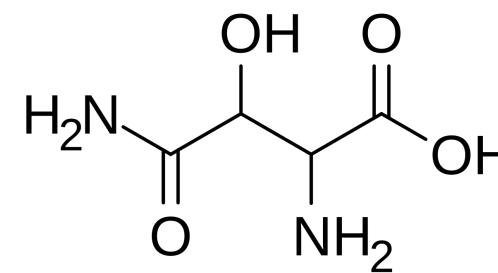
Clovibactin from unculturable bacteria



Multi-target mechanism accounts for lack of resistance



The researchers used biochemical assays, solid-state nuclear magnetic resonance, and atomic force microscopy, to dissect Clovibactin's mode of action and found that it has an unusual killing mechanism. It targets not just one, but three different precursor molecules that are all essential for the construction of the cell wall.



Chemical Signaling

There are four categories of chemical signaling found in multicellular organisms: paracrine signaling, endocrine signaling, autocrine signaling, and direct signaling across gap junctions

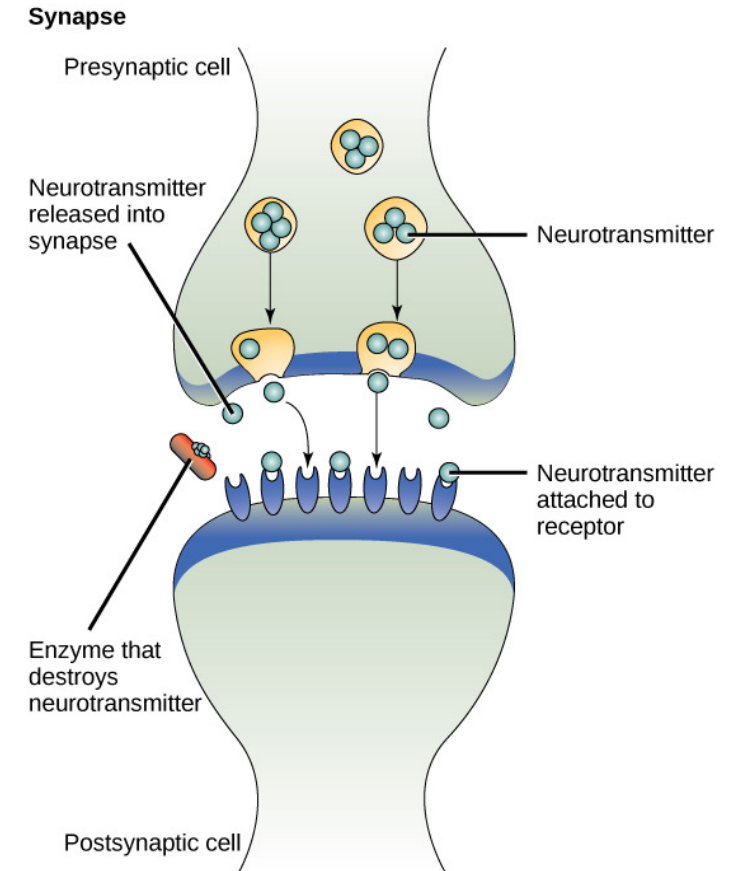
Signals that act locally between cells that are close together are called **paracrine signals**.

Signals from distant cells are called **endocrine signals**, and they originate from **endocrine cells**.

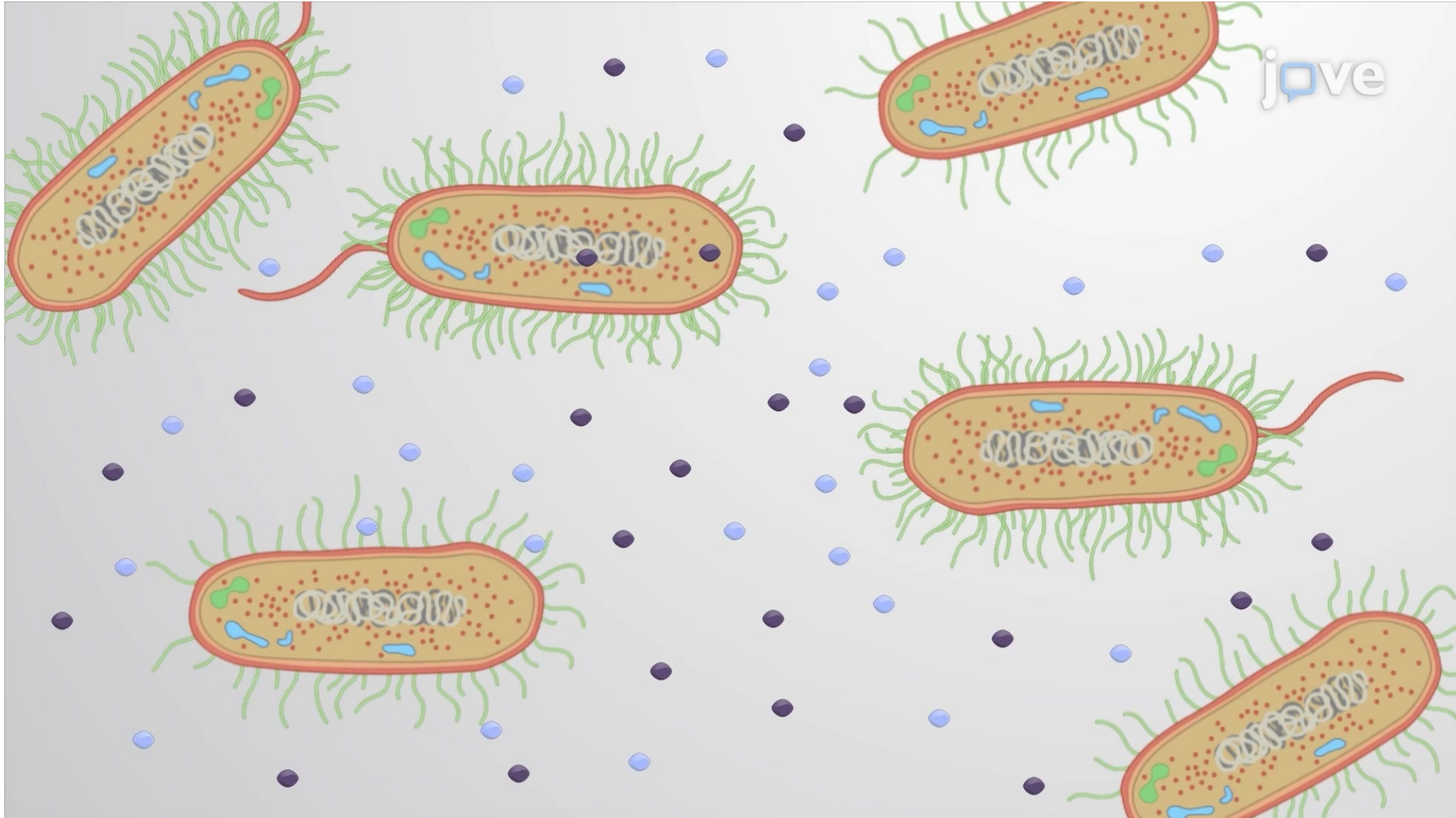
Autocrine signals are produced by signaling cells that can also bind to the ligand that is released. This means the signaling cell and the target cell can be the same or a similar cell (the prefix *auto-* means self, a reminder that the signaling cell sends a signal to itself).

Direct Signaling Across Gap Junctions

Gap junctions in animals and plasmodesmata in plants are connections between the plasma membranes of neighboring cells. These water-filled channels allow small signaling molecules, called **intracellular mediators**, to diffuse between the two cells.

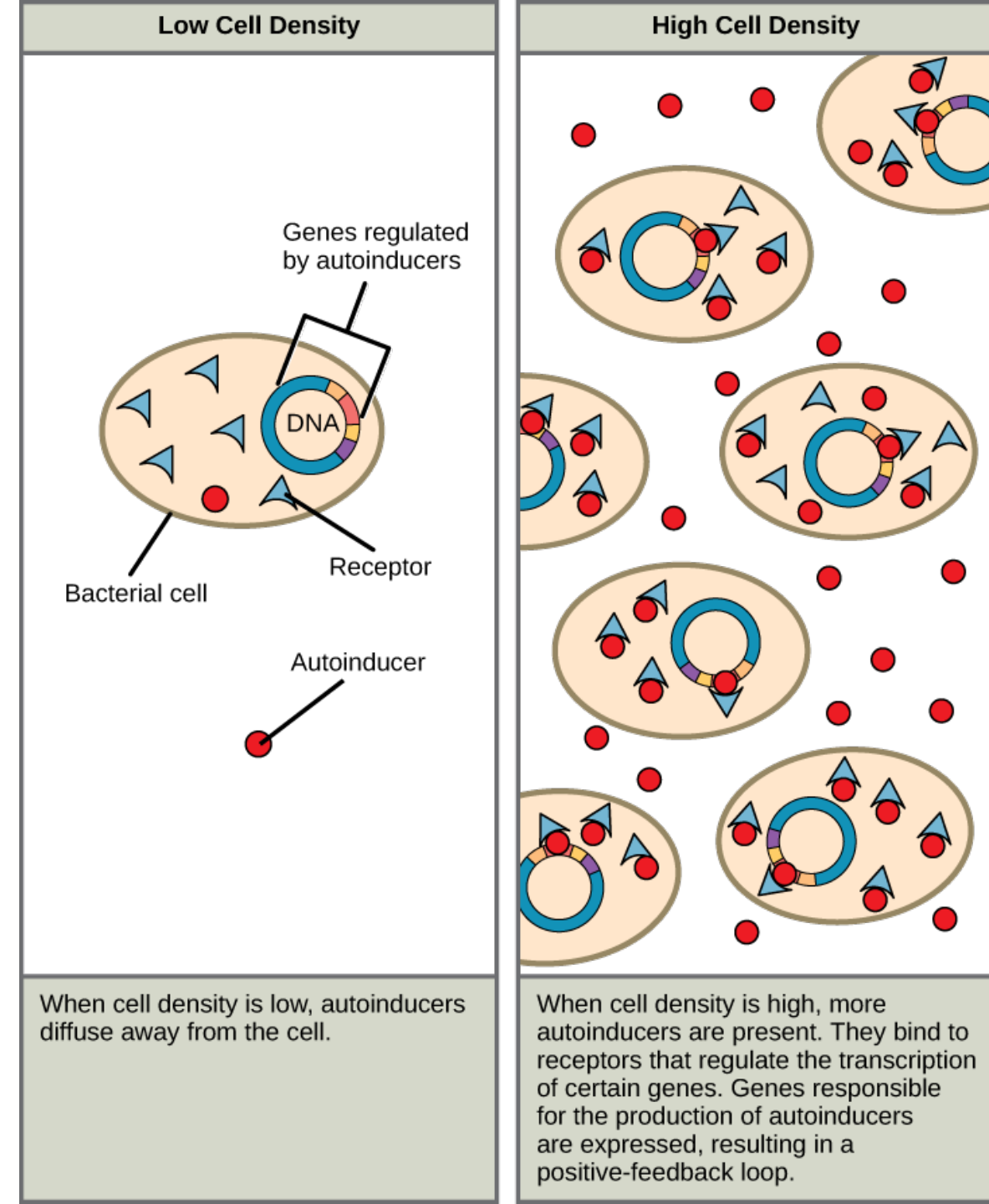


Chemical Signaling in Bacteria



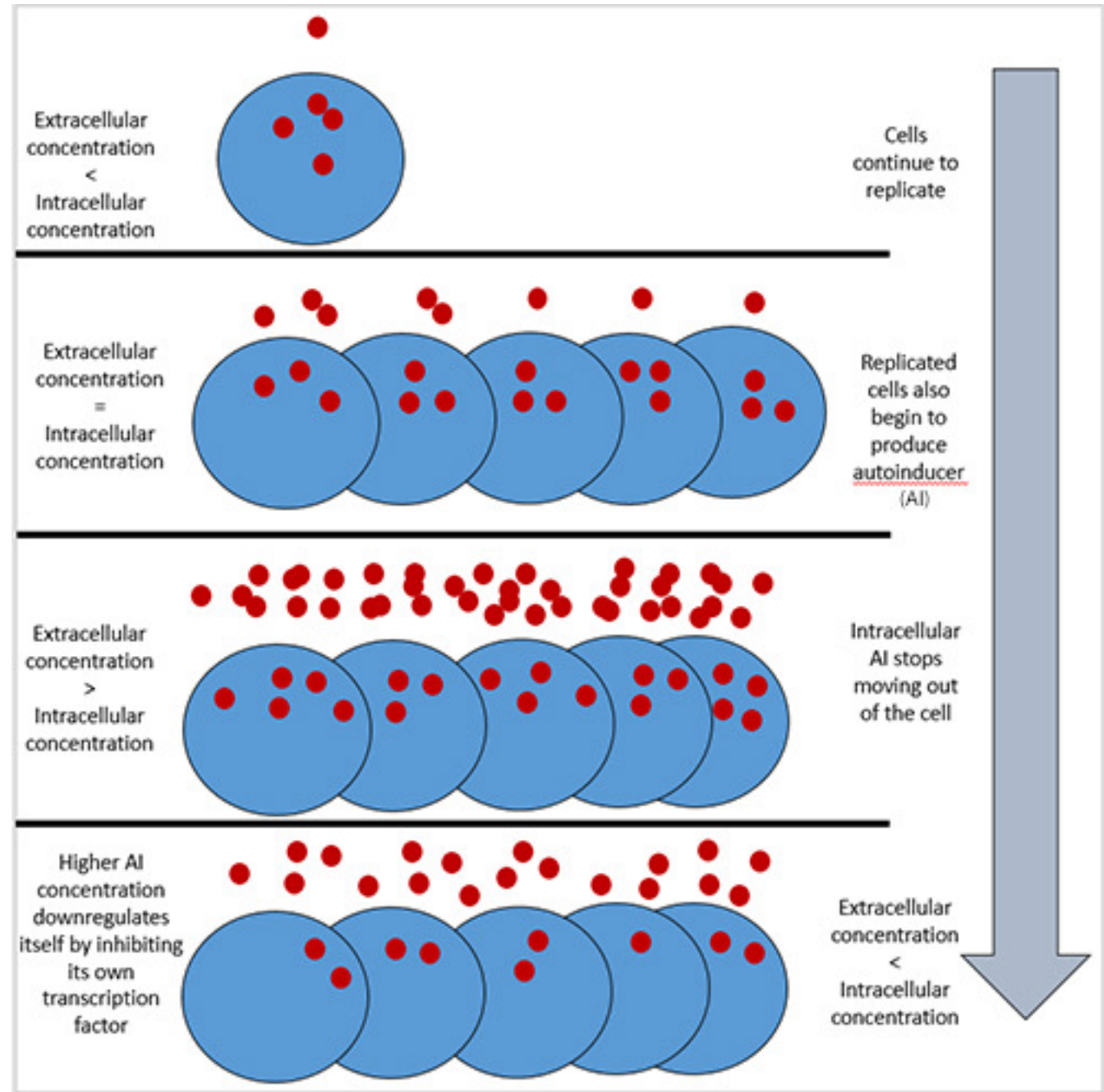
Bacteria employ numerous strategies to sense their local environment and integrate this spatial information to regulate their growth, physiology, and behaviors. **Quorum Sensing** is a form of cell-to-cell communication that enables bacterial populations to coordinate their behaviors in an environmental and cell density-dependent manner

Quorum sensing uses autoinducers as signaling molecules. **Autoinducers** are signaling molecules secreted by bacteria to communicate with other bacteria of the same kind. The secreted autoinducers can be small, hydrophobic molecules such as acyl-homoserine lactone, (AHL) or larger peptide-based molecules; each type of molecule has a different mode of action. When AHL enters target bacteria, it binds to transcription factors, which then switch gene expression on or off.

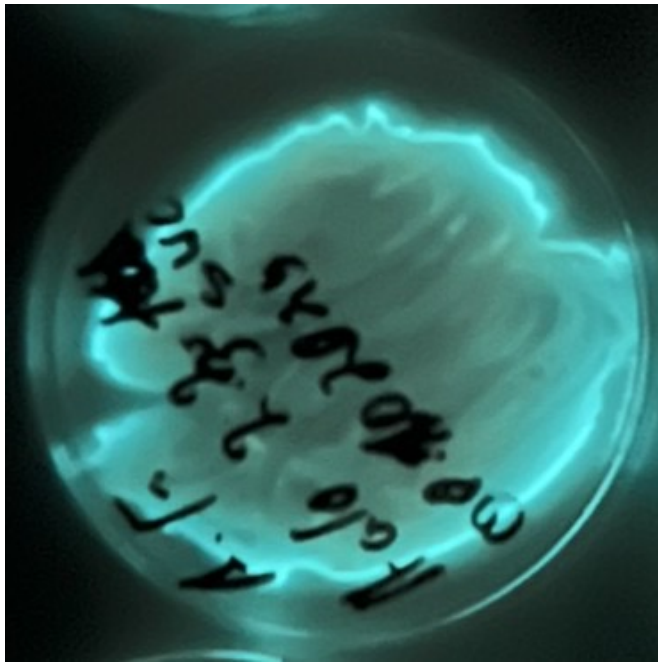


During their reproductive cycle, individual bacterium synthesize autoinducers. Gram-negative bacteria produce N- [acyl-homoserine lactone autoinducers](#) that can passively diffuse through their thin cell wall. In contrast, gram-positive bacterial autoinducers are made of peptide and must be [actively transported](#) through their peptidoglycan cell wall using the [ATP-binding cassette \(ABC\) transporter system](#). In both cases, autoinducers move out of individual cells as they are produced. Since the bacteria are reproducing, there are progressively more individual cells producing autoinducers and the extracellular concentration of the autoinducers increases, eventually hitting a “critical mass”.

Once intracellular concentration increases, autoinducers bind to their receptors, triggering signaling cascades that alter transcription factor activity and therefore, gene expression..



Als	AHL signals	Peptide signals
Are they peptides?	 No	 Yes
Chemical nature	Small organic molecules	Short chains of amino acids
Basic structure	Lactone ring + fatty acyl chain	Linear or cyclic peptides
Typical bacteria	Gram-negative	Gram-positive
How they move	Diffuse across membrane	Actively secreted
Receptor location	Inside the cell	On the cell surface
Example	Acyl-homoserine lactone	Autoinducing peptide (AIP)

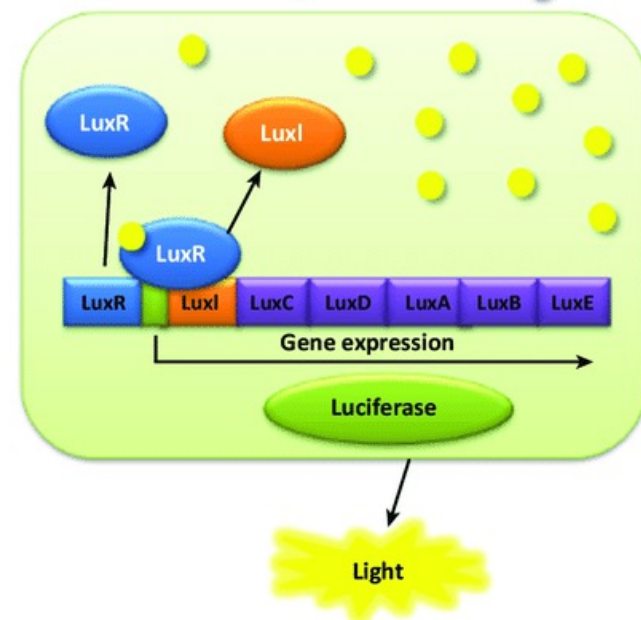
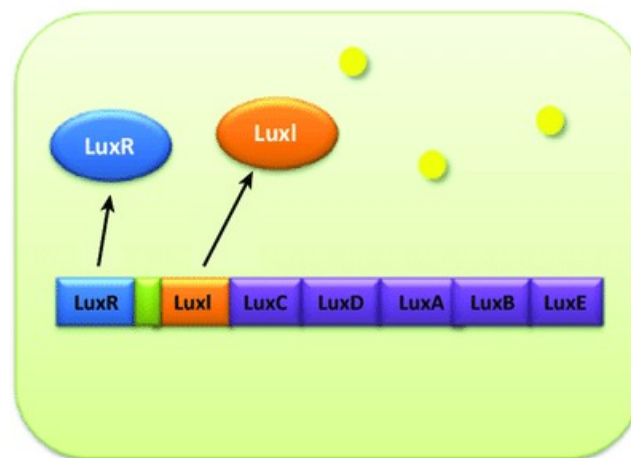
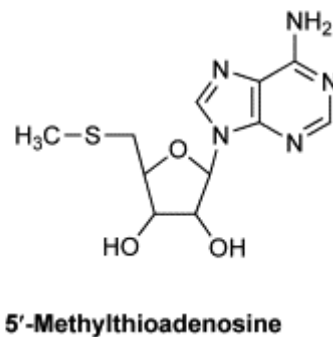
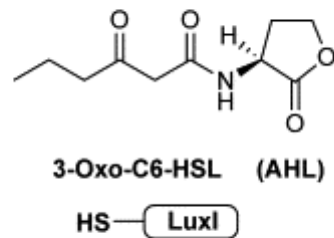
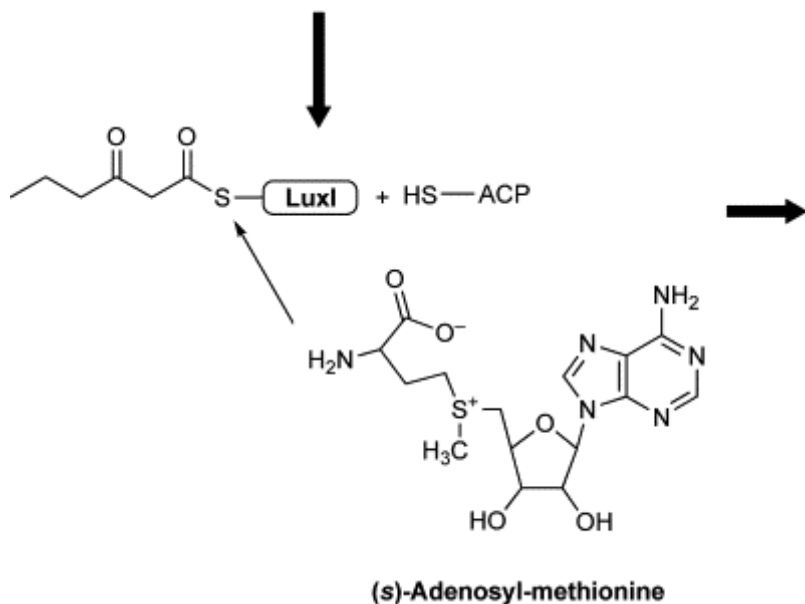
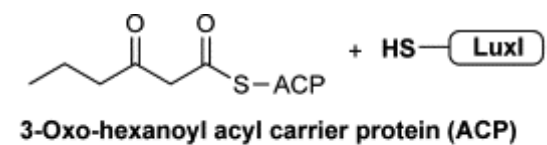


Allivibrio fischeri



Bobtail Squid

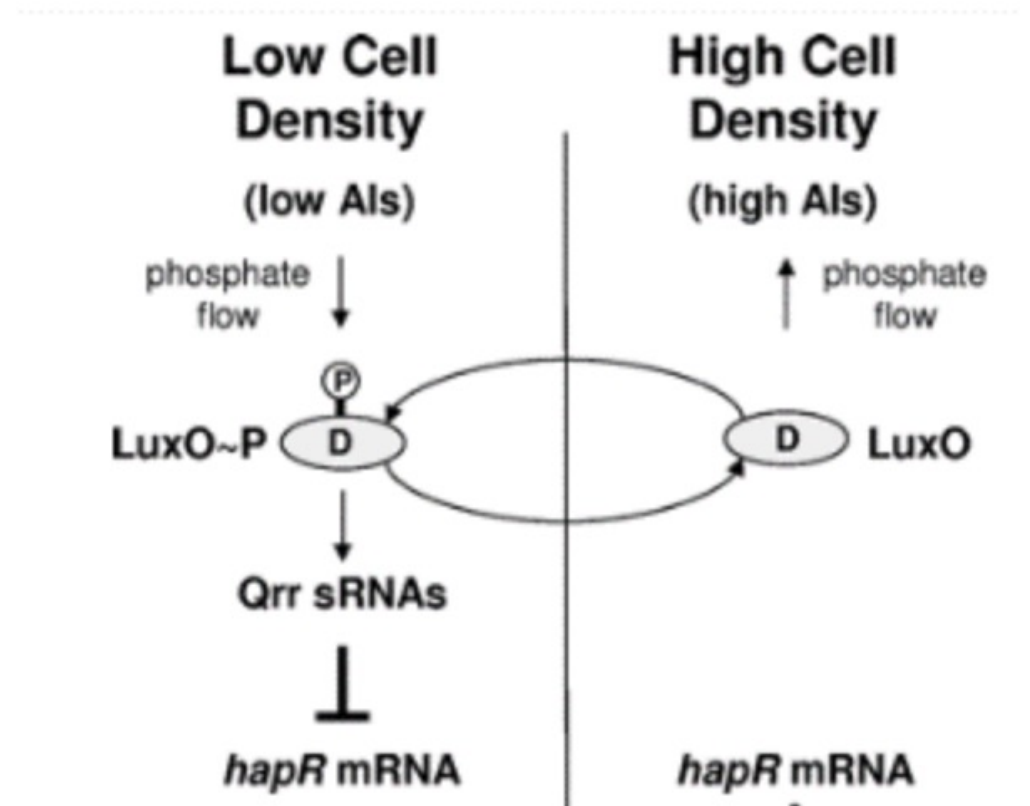
The bacteria reside inside a structure on the underside of the squid called the light organ. At night, the bacteria produce a glow that matches the moonlight coming from above, rendering the squid virtually invisible to predators from below.



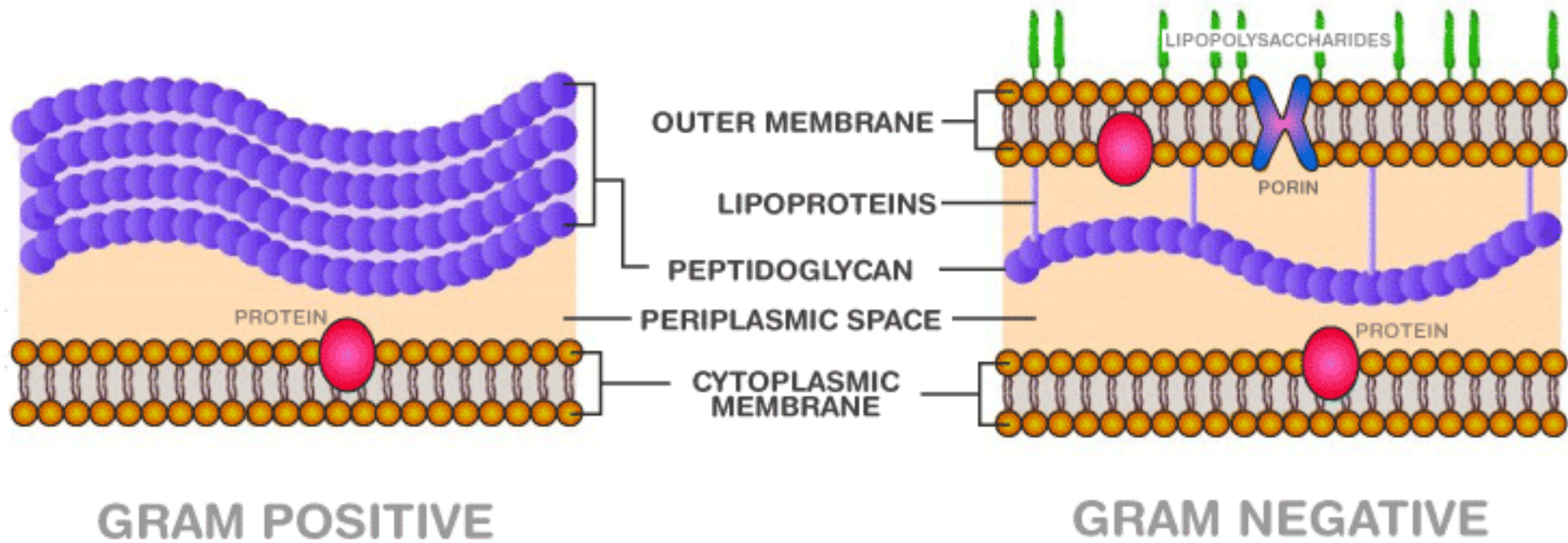
The infamous gram-negative pathogenic bacteria *Vibrio cholerae* uses quorum sensing for virulence during a cholera infection. *V. cholerae* builds biofilms to help transport nutrients between colonies while simultaneously protecting them. The ability to form biofilms within its host ensures the success of the bacteria's reproduction cycle and eventual secretion of cholera toxin, that contribute to [21,000 to 143,000 cholera deaths worldwide](#) each year.

V. cholerae regulates its response to autoinducers via LuxO. When the intracellular concentration of autoinducers is low, [autoinducer receptors act as kinases](#), transferring phosphate to LuxO (LuxO~P). LuxO~P promotes the expression of proteins used for building biofilms.

As *V. cholerae* reproduces, the extracellular concentration of autoinducers increases and eventually exceeds each bacterium's intracellular autoinducer concentration. Intracellular autoinducer concentration in turn increases, [autoinducers bind their receptors and change receptor activity from kinase to phosphatase](#). The receptors then dephosphorylate LuxO, leading to shut down of biofilm-specific gene expression.

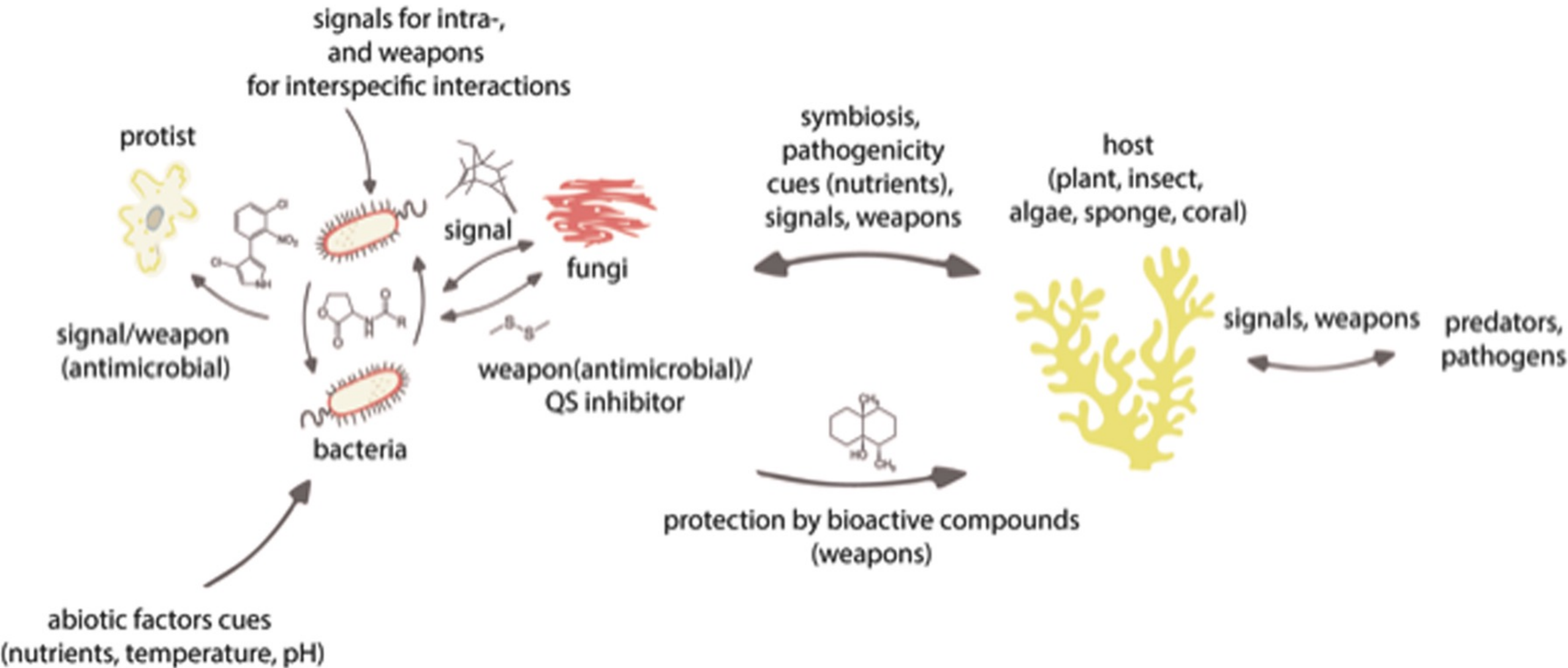


GRAM-POSITIVE AND GRAM-NEGATIVE BACTERIA

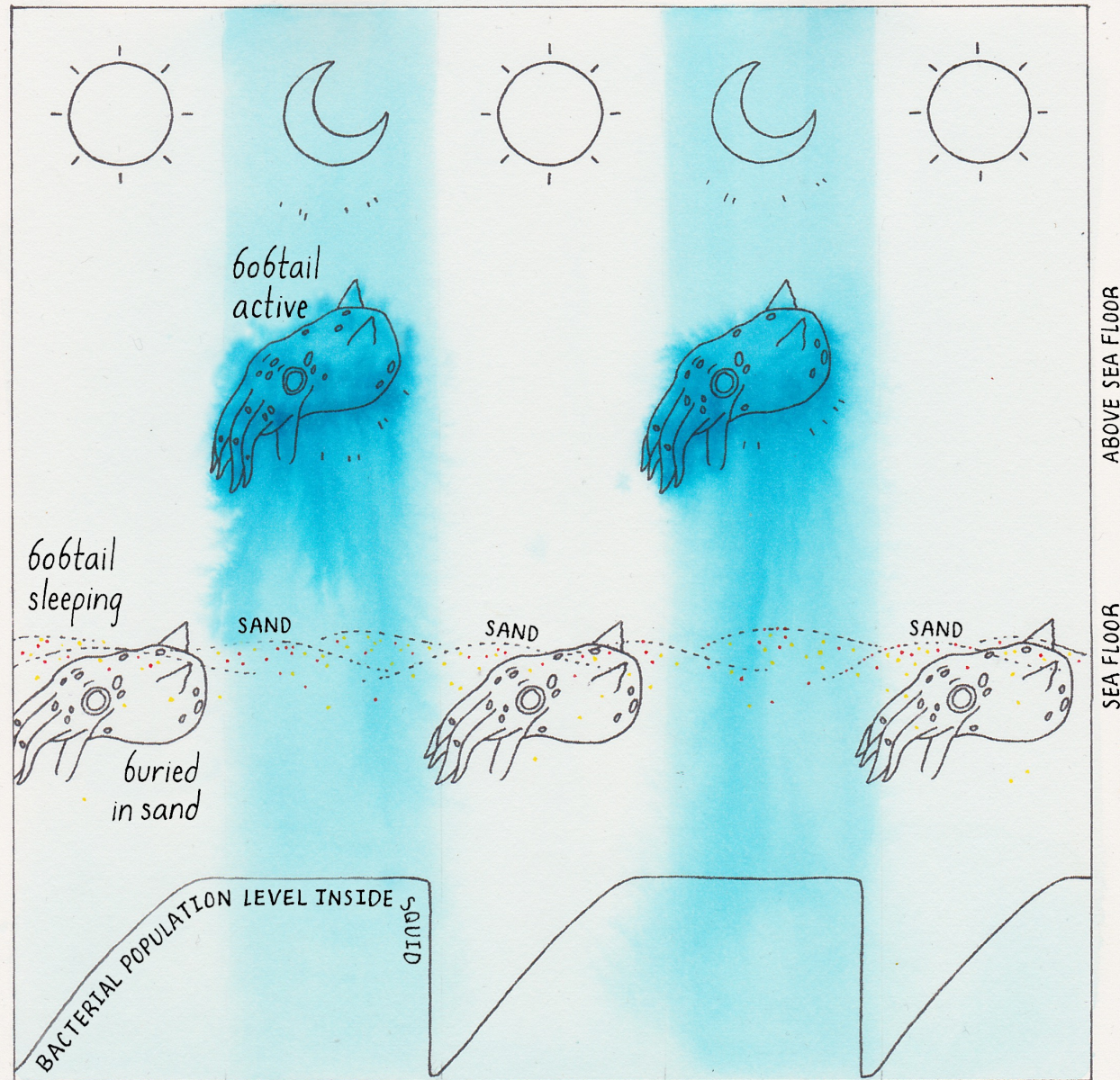


Microbial Signaling

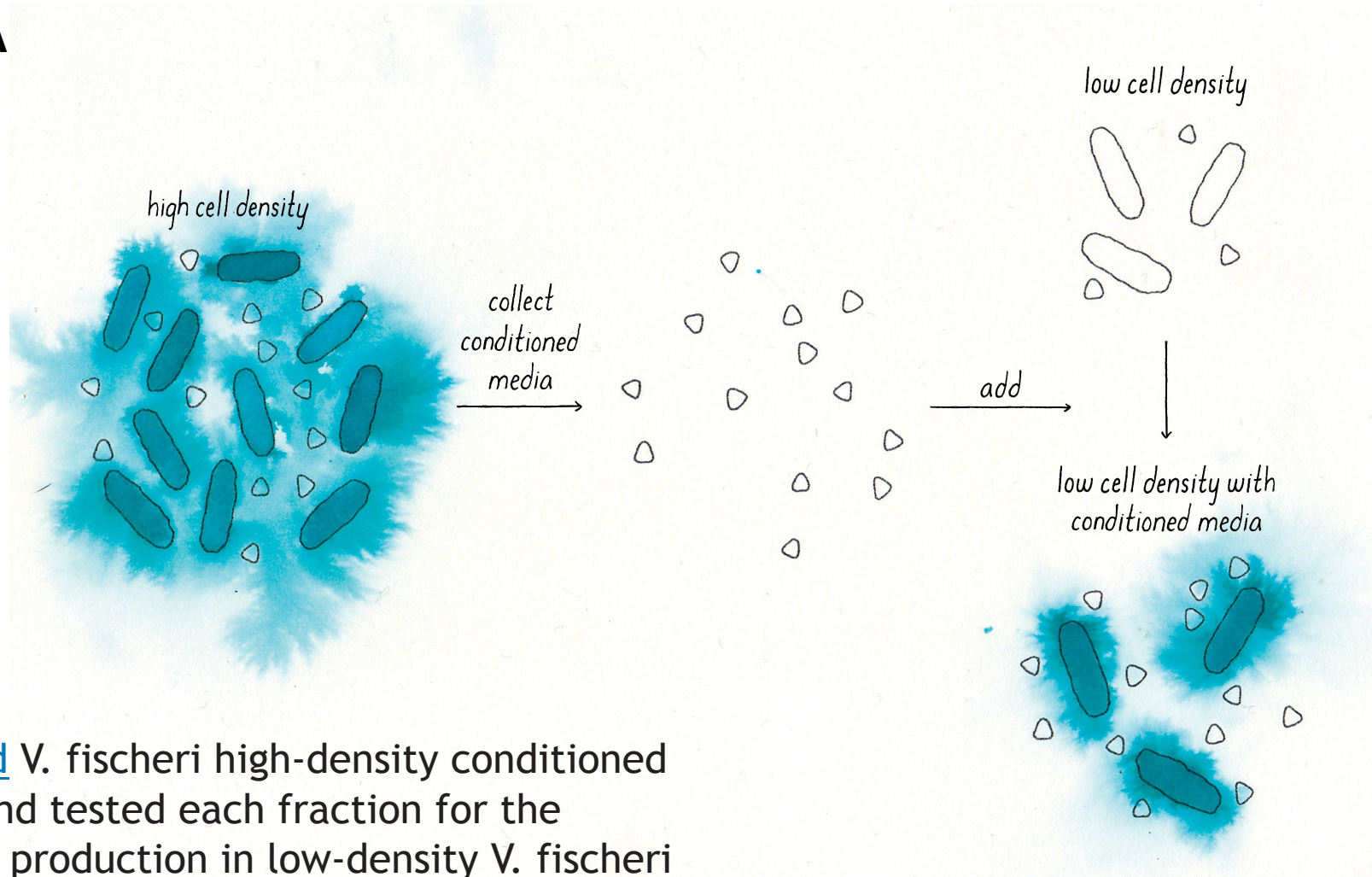
Bacteria communicate with one another—not with words but with chemicals.



BOBTAIL SQUID & BACTERIAL SYMBIOSIS



How to identify the A



Eberhard [fractionated](#) *V. fischeri* high-density conditioned media preparations and tested each fraction for the ability to induce light production in low-density *V. fischeri* cells. Tracking the activity by this method allowed him to isolate the compound, and, using companion chemical analyses, Eberhard determined its structure. In this first case, he identified the molecule as a compound called a homoserine lactone.

The identification of the autoinducer was important, but how did it work? That question was tackled by Mike Silverman and his graduate student Joanne Engebrecht in 1983.

Engebrecht and Silverman designed a cunning but fantastically simple experiment to identify the components that enabled *V. fischeri* to glow.

Their strategy: take DNA from *V. fischeri* and transfer pieces of it into *E. coli* and then check for glowing *E. coli*, presumably containing a piece of *V. fischeri* DNA with important genes required for light production.

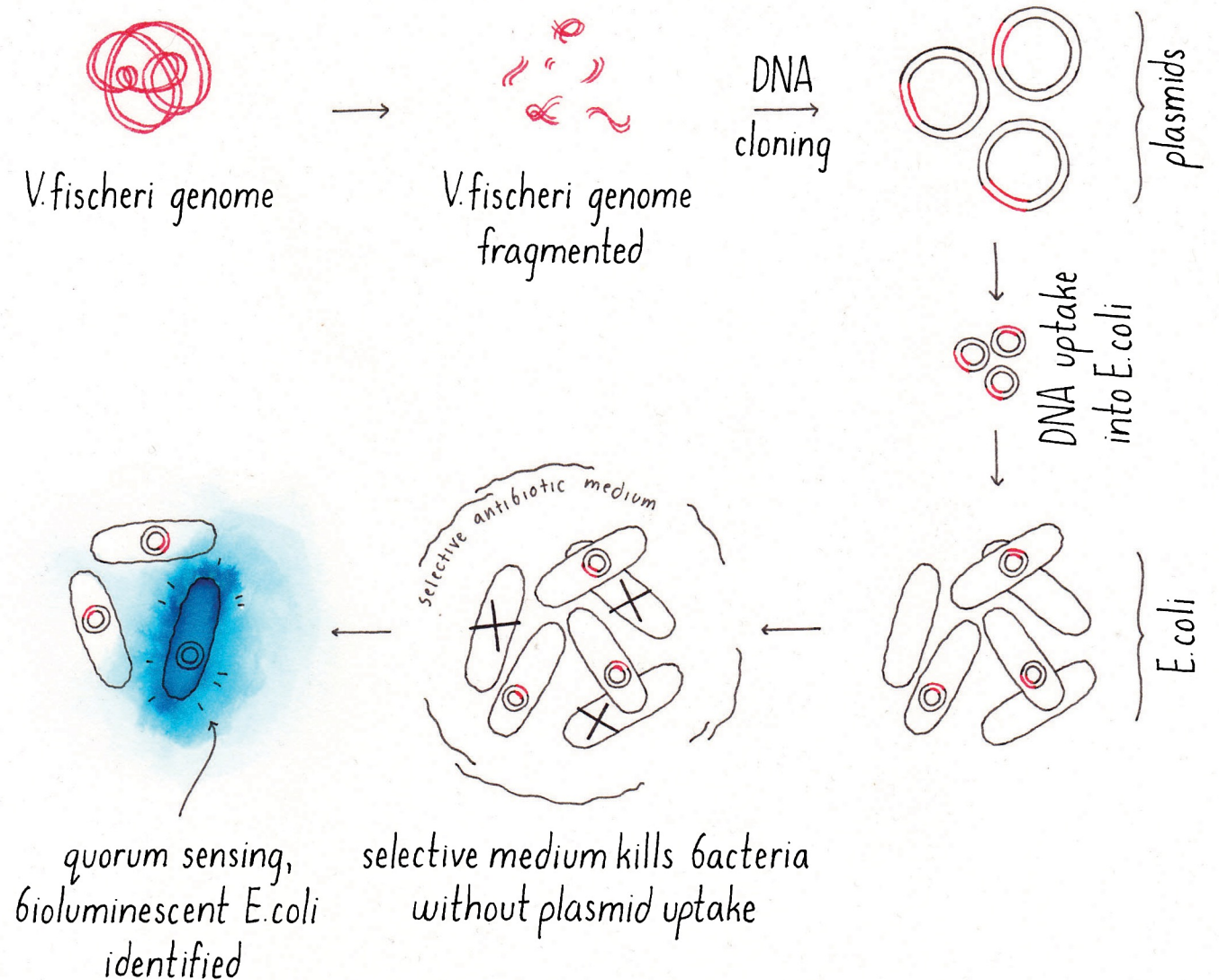
Of course, they did not know which *V. fischeri* genes were responsible for bioluminescence itself, for making the autoinducer, or for the response to the autoinducer. That was the point of the experiment.

So, they sheared the *V. fischeri* genome into fragments and cloned all of the segments into plasmids. They introduced this “library” of plasmids (each plasmid containing a random fragment of *V. fischeri* genomic DNA) into *E. coli*. They grew the *E. coli* into colonies on Petri plates, turned the lights off in the room, and in the darkness, they scanned their Petri plates for a recombinant *E. coli* colony that glowed in the dark

Remarkably, Englebrecht and Silverman found an *E. coli* colony that glowed in the dark! They isolated the fragment of *V. fischeri* DNA from the glowing *E. coli* colony and sequenced it to identify the genes contained on the DNA fragment. This analysis showed that the DNA fragment contained several genes encoding the following proteins: (1) the enzyme that makes the autoinducer, (2) an autoinducer receptor (the protein that binds and detects the autoinducer), and (3) the proteins that make light.

They showed that their bioluminescent *E. coli* produced the autoinducer and released it into the culture fluids and only emitted bioluminescence at high cell density. Thus, in one elegant experiment, and on one single fragment of DNA, they identified the genes for cell-cell communication, and the genes for light production and the entire system functioned properly in *E. coli*.

THE ENGELBRECHT & SILVERMAN EXPERIMENT



Englebrecht, J. K. Nealson, and M. Silverman. Bacterial bioluminescence: Isolation and genetic analysis of functions from *Vibrio fischeri*. *Cell* 1983;32:773-781.

Silverman named the genes required for light production “Lux,” for Luminescence expression with the idea that people would be reminded of Luxor the Egyptian city housing the temple for the God of Light. Engebrecht and Silverman came up with “LuxI” as the name for the “Inducer” (more accurately LuxI is the enzyme that synthesizes the autoinducer), and “LuxR” for the “Receptor” that binds the autoinducer. Engebrecht and Silverman proposed a brilliant model



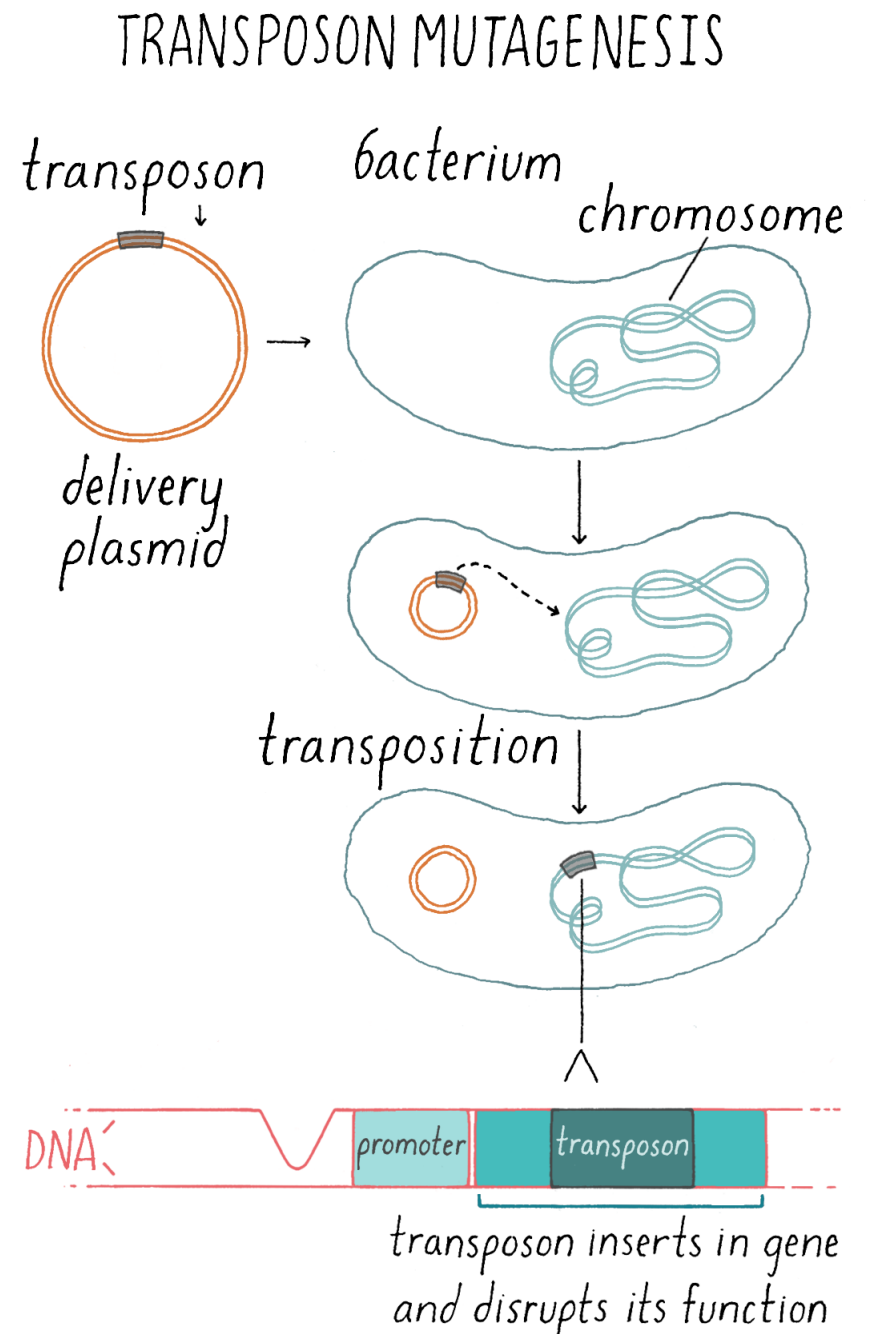
Silverman and Bassler wanted to find out if there were other, interesting components involved in quorum sensing that were not discovered in his original experiment. They wanted to do the opposite: take bacteria that normally glowed in response to autoinducers and introduce mutations that would cause them to stop glowing and reasoned that they might be able to find additional genes by this method.



Vibrio harveyi, a bioluminescent marine bacterium that is closely related to *V. fischeri*. *V. harveyi* exists free-living in the ocean and apparently has no symbiotic hosts. However, we knew that *V. harveyi*, like *V. fischeri*, only produced light at high cell density,

if a transposon landed in any of the AI genes, then that *V. harveyi* mutant would be dark (make no light). They were especially interested in identifying luxI and luxR, thinking that, after they found and characterized those known genes, they could dispense with them and then move on to find out if other interesting components were involved.

The researcher made one overarching assumption, which turned out to be COMPLETELY WRONG!



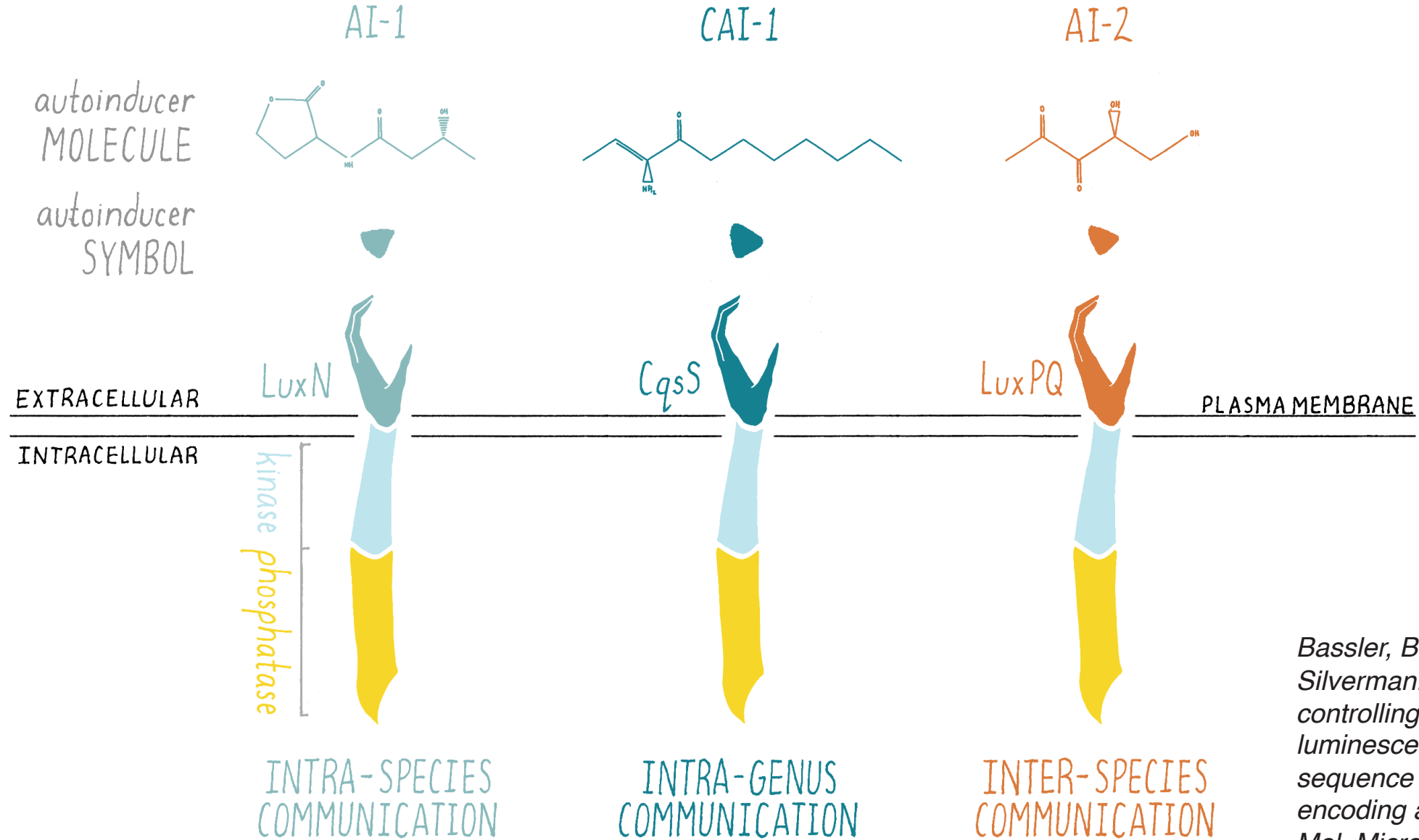
I made transposon insertions and collected non-light-producing mutants for a year. My characterizations, over and over, showed that all the genes that had transposon insertions were genes encoding luciferase but never, ever in any other gene, such as luxI or luxR, which I expected would regulate luciferase production.

My results made no sense—I knew there was an autoinducer so there had to be a gene for an enzyme that made it. I knew the autoinducer was detected only at high cell density so there had to be a gene encoding a receptor protein. But I did the same experiment over and over and over. Finally, one day, after I again got the same results, I stopped just doing the experiment that was obviously not working, and sat down to think- why I might not be getting mutations in regulatory genes using this strategy?

the “aha moment” finally came.. When the researcher decided to look for mutants that were dim but not dark (non-light producing). Dim mutants were obviously capable of making some light, so those mutants could not be defective in luciferase genes (luciferase genes are absolutely required for light). However, as such mutants did not make the correct amount of light, it is likely that they might be defective in the proper regulation of light production.

Apparently, *V. harveyi* had several communication systems functioning in parallel.

V. HARVEYI'S MULTIPLE AUTOINDUCERS & RECEPTORS FOR QUORUM SENSING

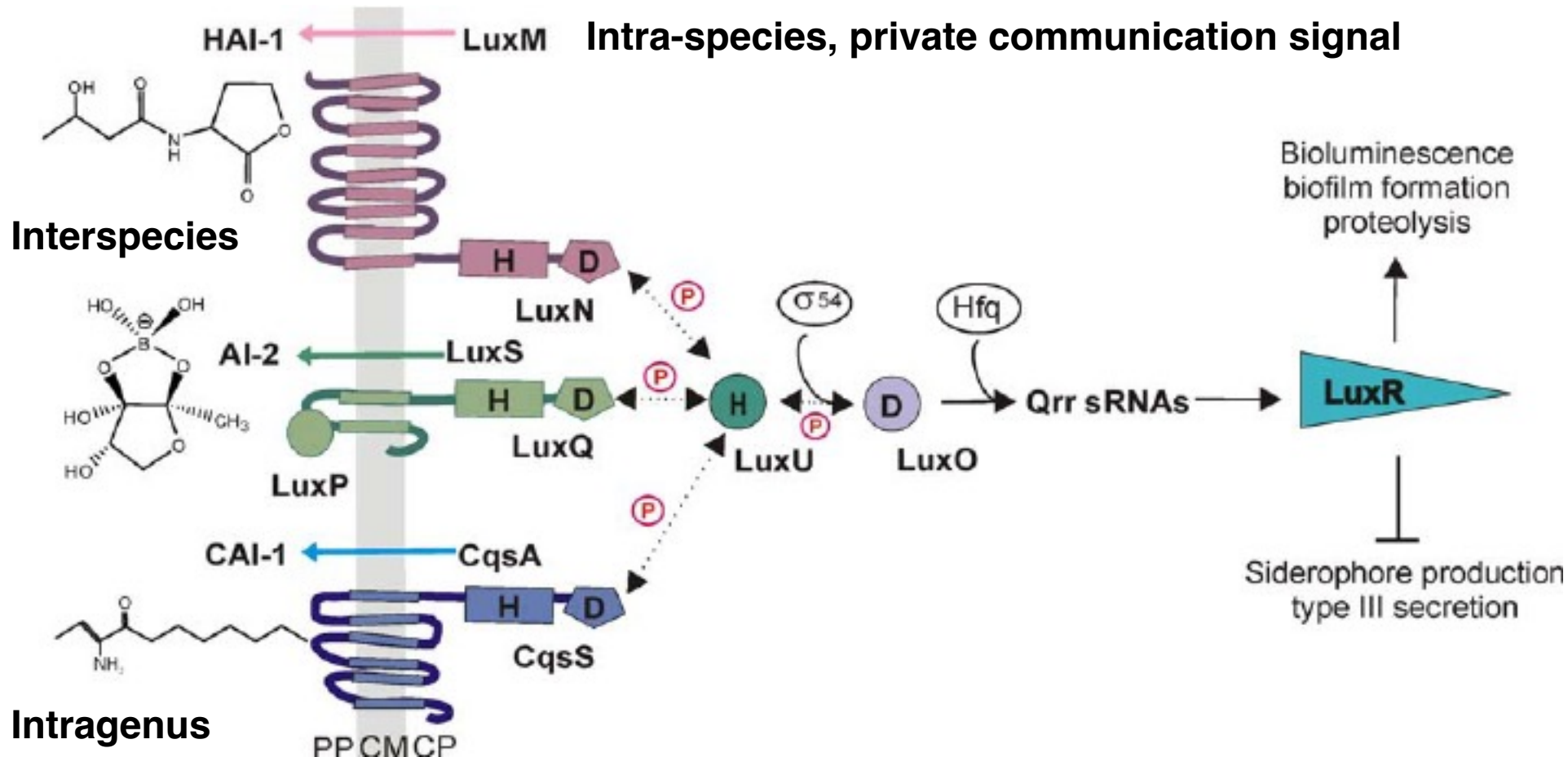


Bassler, B.L., M. Wright, M.R. Silverman. Multiple signaling systems controlling expression of luminescence in *Vibrio harveyi*: sequence of function of genes encoding a second sensory pathway. *Mol. Microbiol.* 1994;**13**:273–286.

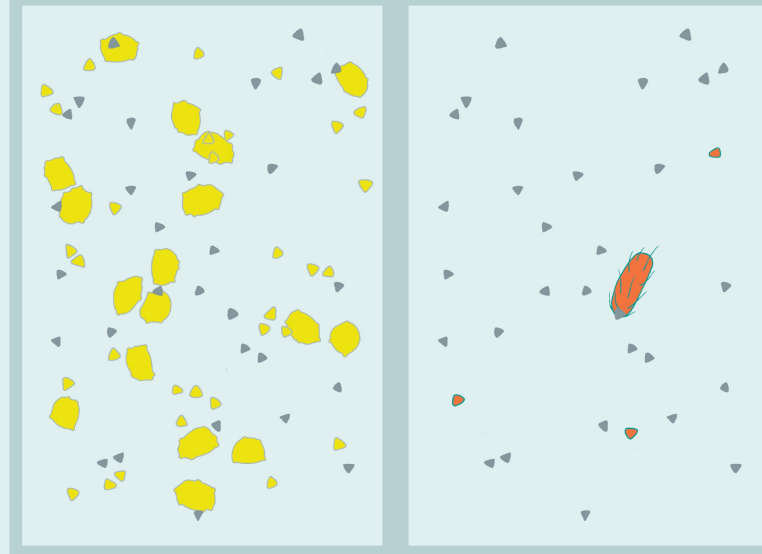
Intra-species, Intra-genera, and Inter-species Communication

Bacteria often use multiple autoinducers for quorum sensing and the chemical nature of each autoinducer encodes different information about “who” is in the vicinity.

V. harveyi uses three different/parallel autoinducer chemicals for intra-species, intra-genera, and inter-species communication, respectively



Here, there are many “yellow” bacteria in the community. They encounter a high level of their own intra-species autoinducer (yellow triangles) as well as a high concentration of the inter-species autoinducer (blue triangles). That blend of autoinducer tells the “yellow” species that they are at high cell density but they have neighbors that are non-kin.



cell

- ▲ respective intraspecies autoinducer
- ▼ general interspecies autoinducer

$$\frac{\text{INTRASPECIES}}{\text{INTERSPECIES}} = \text{HIGH}$$



cell

- ▲ respective intraspecies autoinducer
- ▼ general interspecies autoinducer

$$\frac{\text{INTRASPECIES}}{\text{INTERSPECIES}} = \text{LOW}$$

Here, there are only a few bacteria of the “orange” species present. The “orange” bacterial species detects none or only a very low level of its species-specific autoinducer (orange triangles) but it encounters a high level of the interspecies autoinducer (blue triangles). That combination of autoinducers tells the “orange” species that it and its kin are far outnumbered by non-kin.

- Certain bacteria secrete enzymes that can clip a competitor's autoinducer in half, making it inactive.
- Other bacteria ravenously consume their competitor's autoinducers, using the chemical building blocks in the autoinducer (carbon, nitrogen) as nutrients to help the consumer to grow.

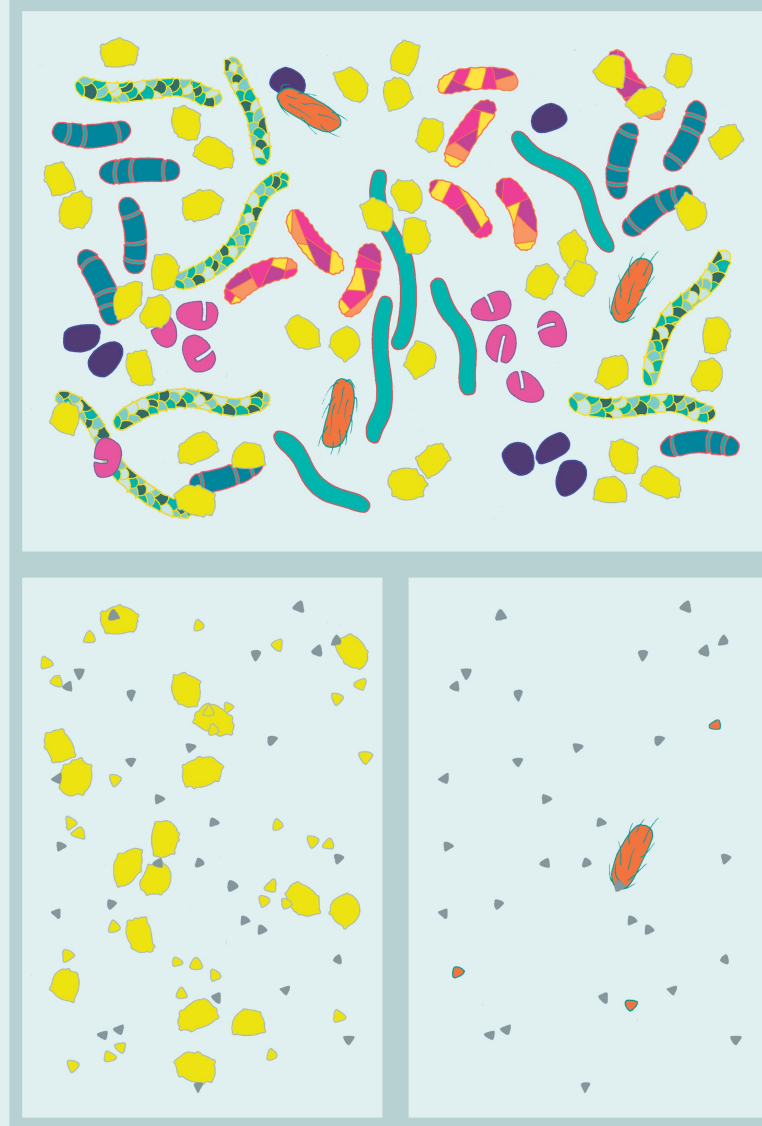
Certain species of *Bacillus* secrete an enzyme (called a lactonase) that cleaves the carbon rings of homoserine lactone autoinducers produced by other bacteria.

Dong et al. introduced the gene that encodes the lactonase AiiA enzyme into plants to make the plants resistant to bacterial pathogens that require quorum sensing to cause disease.



Dong, Y.-H., L.-H. Wang, J.-L. Xu, H.-B. Zhang, X.-F. Zhang, and L.-H. Zhang. Quenching quorum-sensing-dependent bacterial infection by an N-acyl homoserine lactonase. Nature 2001;411:813–817.

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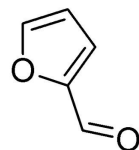
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Microbial volatile compounds

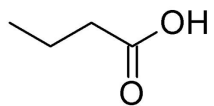
Bacteria and other microorganisms synthesize a huge amount of volatile organic compounds (VOCs). Microbial VOCs belong to different classes of chemical compounds, including alkenes, alcohols, ketones, benzenes, pyrazines, sulfides, terpenes, etc. It was found that many VOCs can modulate (suppress or stimulate) the growth of microorganisms and plants, cause systemic plant resistance, and affect insects, nematodes, and other organisms.

VOC production can play a significant role in the competitive relationships of microorganisms. In addition, VOCs can function as novel communication factors (“**infochemicals**”) between bacteria as well as between microorganisms and higher organisms. They act remotely when transmitted by air (or through the aquatic environment), influencing the expression of genes of target organisms

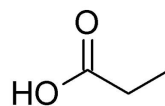
Microbial volatile organic compounds (MVOCs) are a type of VOCs produced by all microorganisms as part of their normal metabolism. The comparative analysis of experimental data has shown that volatile metabolites make a much greater contribution to the microbial interactions than non-volatile ones. It has been found that interaction of microorganisms via the volatiles they release occurs frequently and is typical of a number of microorganisms



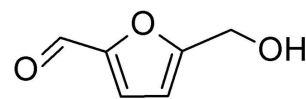
Furfural



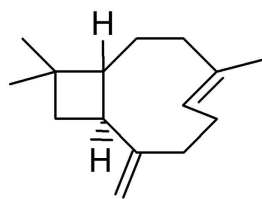
Butanoic acid



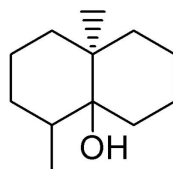
Propanoic acid



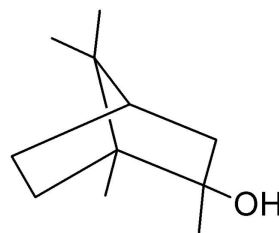
5-Hydroxymethylfurfural



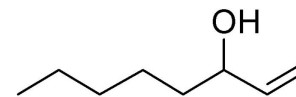
β -caryophyllene



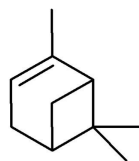
Geosmin



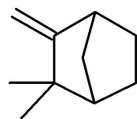
2-Methyl isoborneol



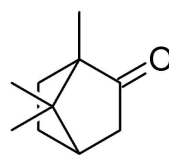
1-octen-3-ol



α -pinene



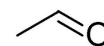
Camphene



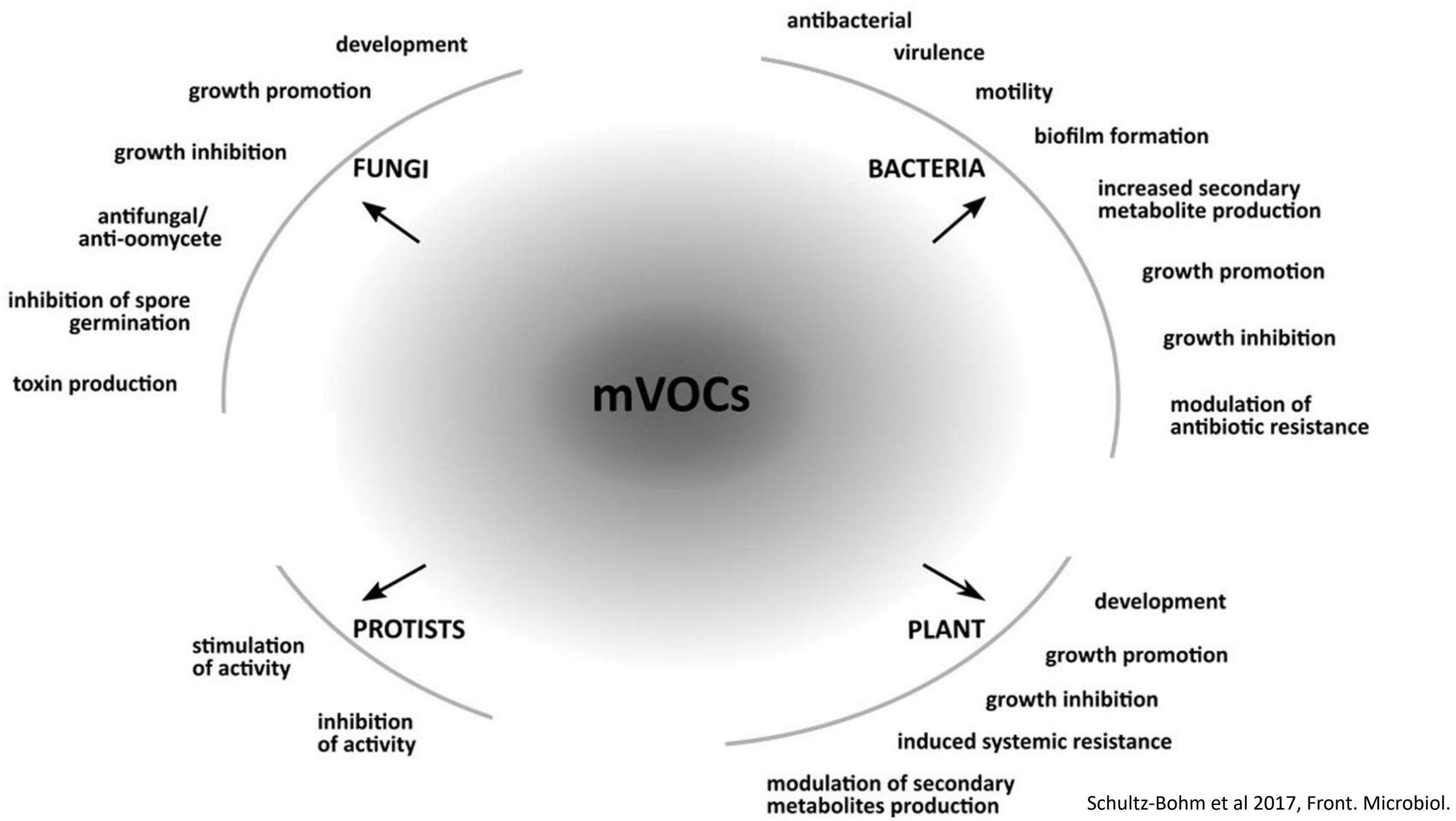
Camphor



Methanol



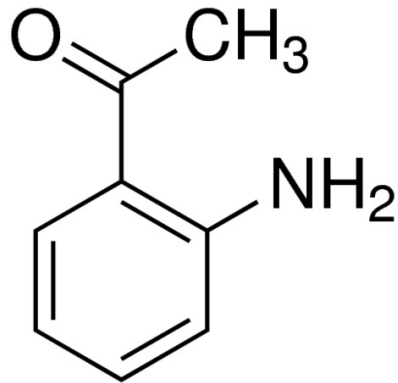
Acetaldehyde



P. aeruginosa cultures are characterized by a sweet grape-like odor. This odor has been attributed to the small-excreted volatile molecule, 2'-amino acetophenone (2-AA), which is used to diagnose *P. aeruginosa* infections in humans

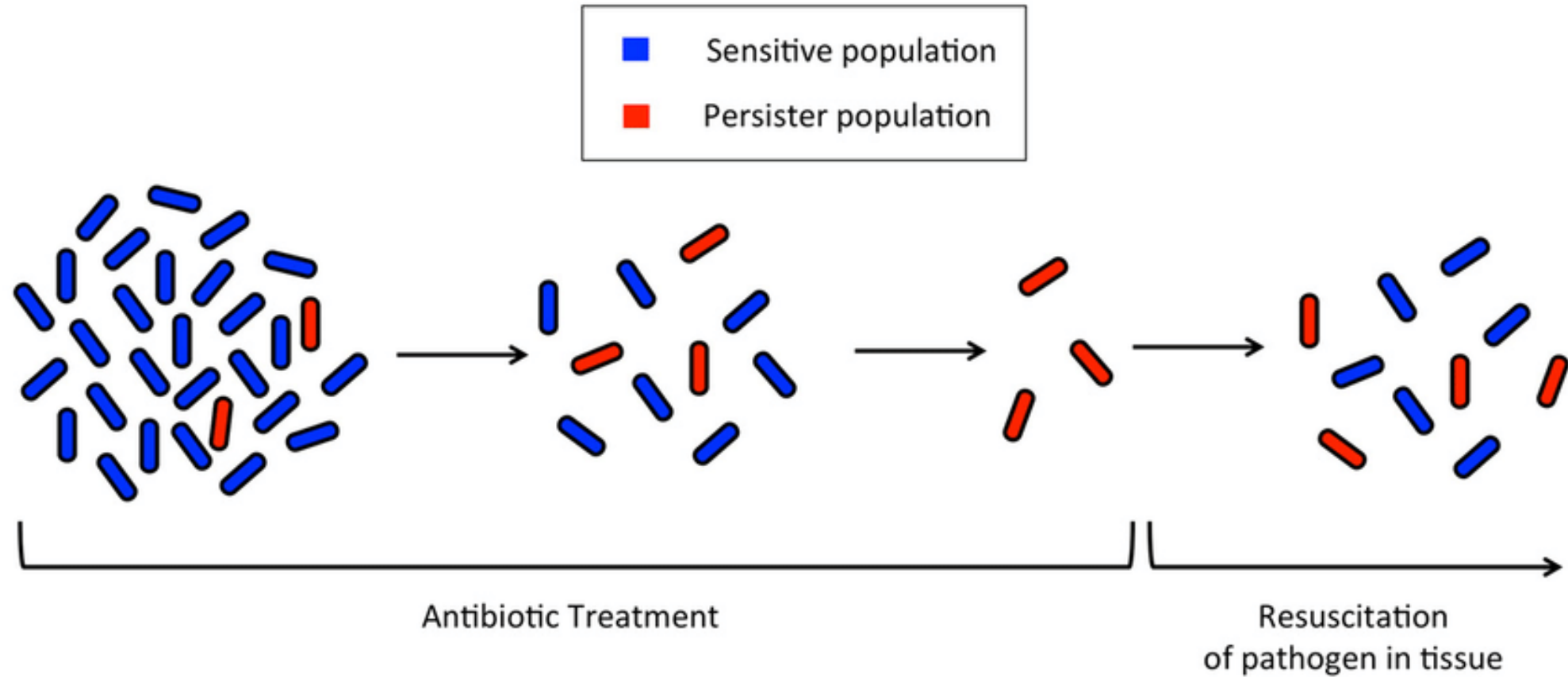
2-AA is usually present in difficult to treat *P. aeruginosa*-infected burn wounds and *P. aeruginosa* clinical isolates from cystic fibrosis patients.

Is it possible that 2-AA may be involved in *P. aeruginosa* antibiotic tolerance?



Pseudomonas aeruginosa (*P. aeruginosa*) is a leading, opportunistic pathogen in burn wound infections, causing significant morbidity and mortality by thriving in moist wounds, forming resilient biofilms and developing multi-drug resistance.

Bacterial Persister Cells

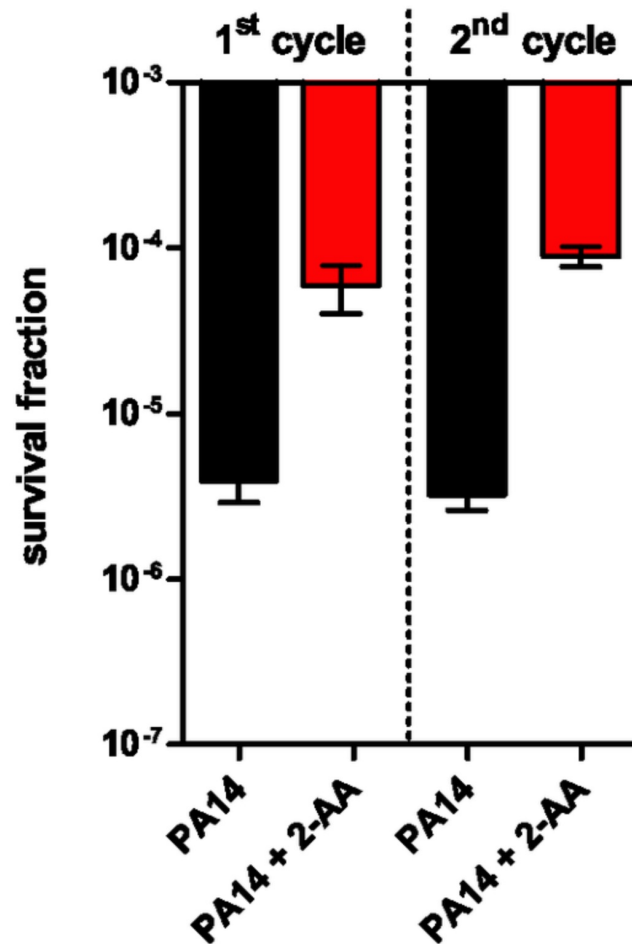
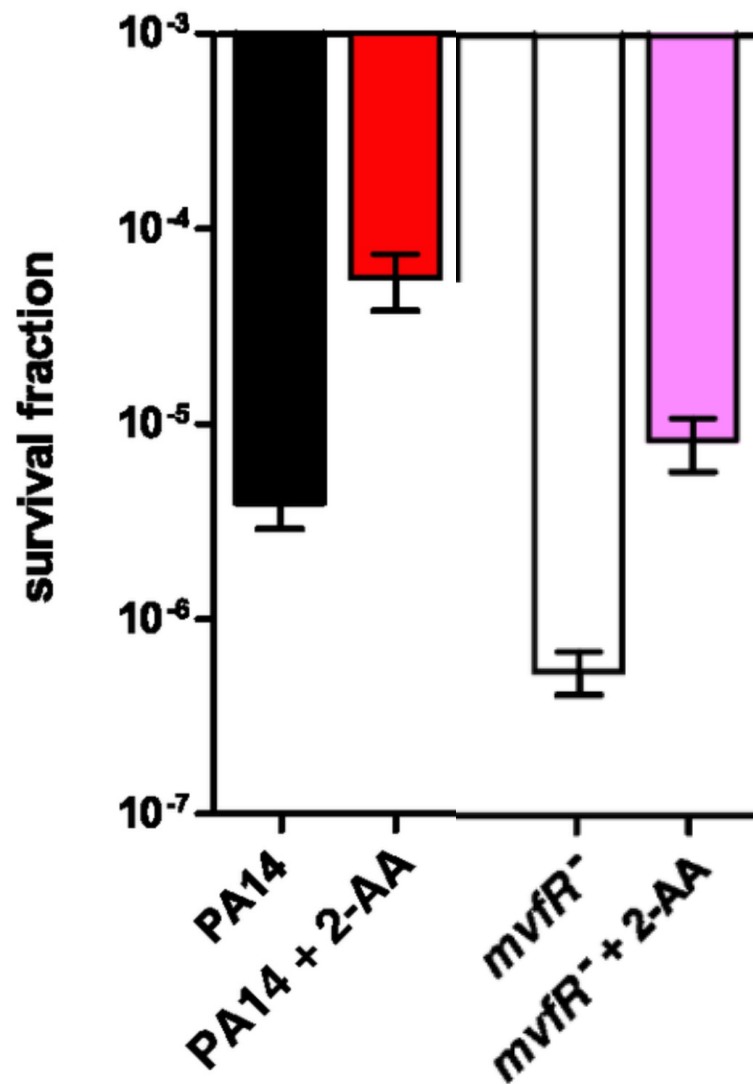


Persister cells are subpopulations of cells that resist treatment, and become **antimicrobial tolerant** by changing to a state of dormancy.

Antibiotic Resistance versus Tolerance

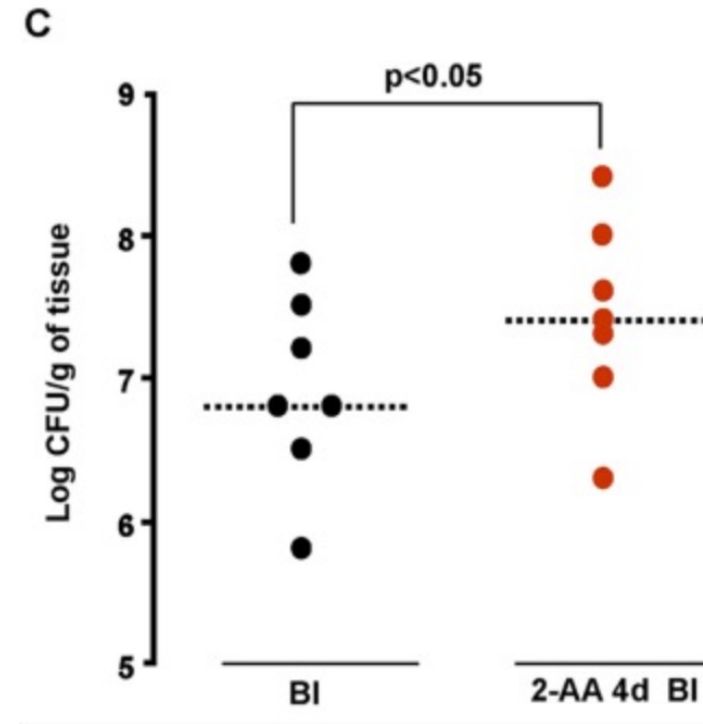
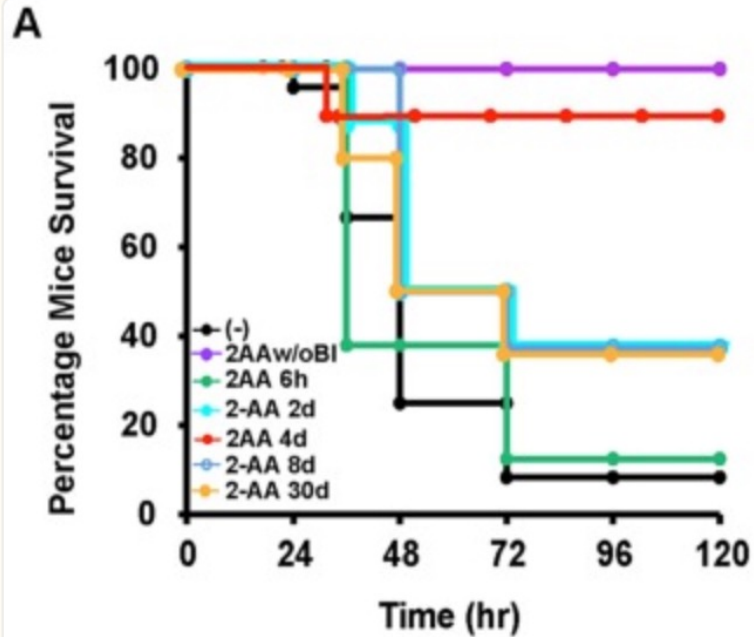
Feature	Resistant Cells	Persister Cells
Basis of survival	Genetic mutation / resistance genes	Phenotypic dormancy
Heritable?	✅ Yes	❌ No
Growth in antibiotics	Continue to grow	Do not grow, but survive
Frequency	Can become dominant	Rare (<1%)
MIC change	Increased	Unchanged
After treatment	Remain resistant	Regrow as susceptible cells
Clinical role	Permanent treatment failure	Relapse & chronic infection

This study provides a direct link among the QS volatile molecule 2-AA and antibiotic tolerance, persister cells, through the modulation of the translational capacity of the cell.



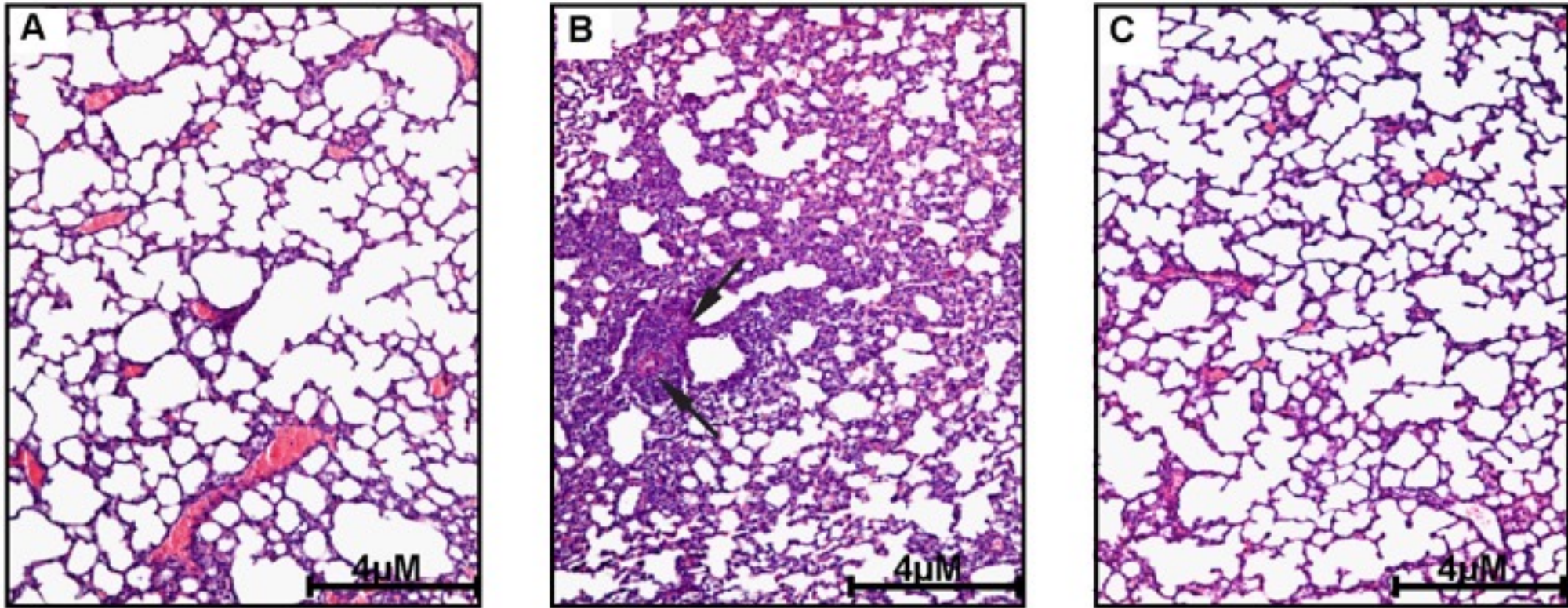
2'-Aminoacetophenone promotes persister formation by reprogramming bacterial metabolism toward a low-energy, stress-tolerant state, thereby increasing survival during antibiotic exposure without inducing genetic resistance.

What does this mean for the host during infection?



- Mice pre-treated with 2-AA **before infection** survive significantly better than untreated mice. ~90% survival with 4-day pretreatment vs ~10% in controls.
- Despite improved survival, **bacterial loads (CFU)** in tissues of 2-AA-treated mice remain **higher** or persist longer than in untreated animals.

2-AA dampens the lethal inflammatory response without clearing bacteria, promoting host *tolerance* and pathogen *persistence*.



(A) Control healthy (non-infected) lung tissue 4 d after 2-AA treatment. (B) Inflammatory cell infiltration with large areas of consolidation in lung parenchyma 48 h after infection with PA14 (Black arrows indicate the infiltration and necrotic foci). (C) Lack of infiltration 48 h after PA14 infection in the lungs of mice pretreated with 2-AA 4 d prior to BI.

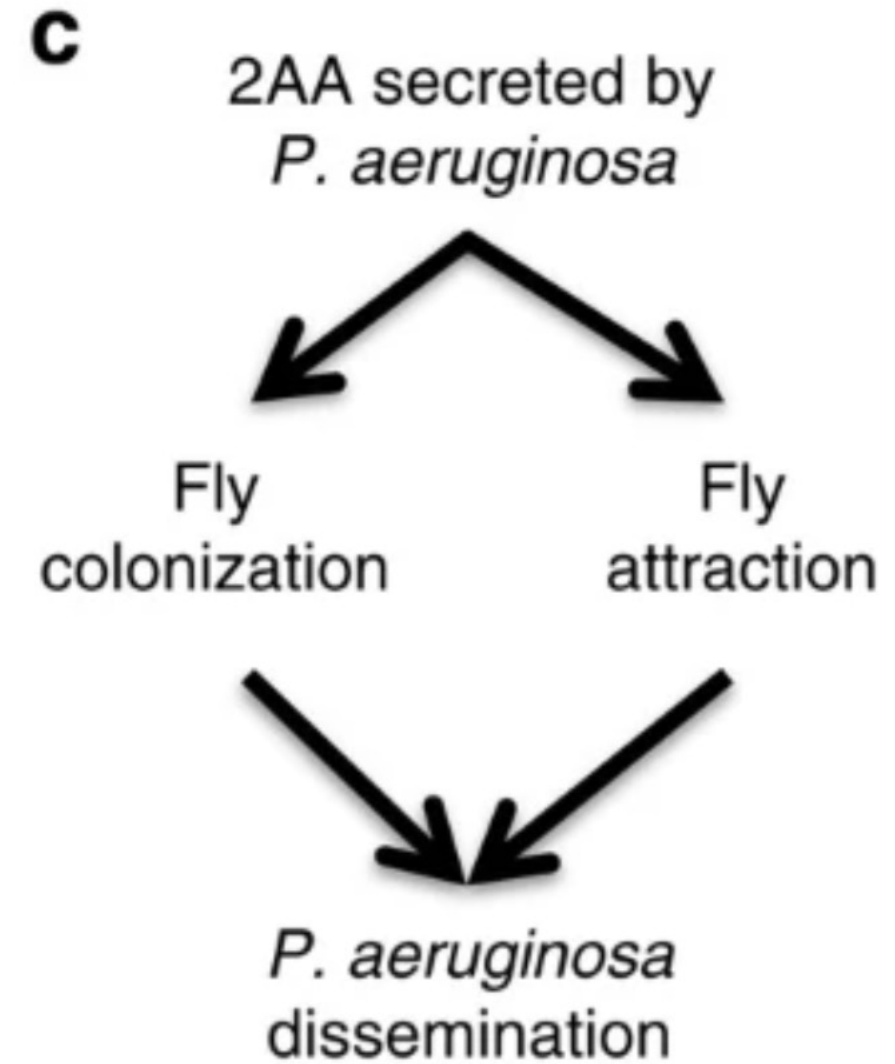
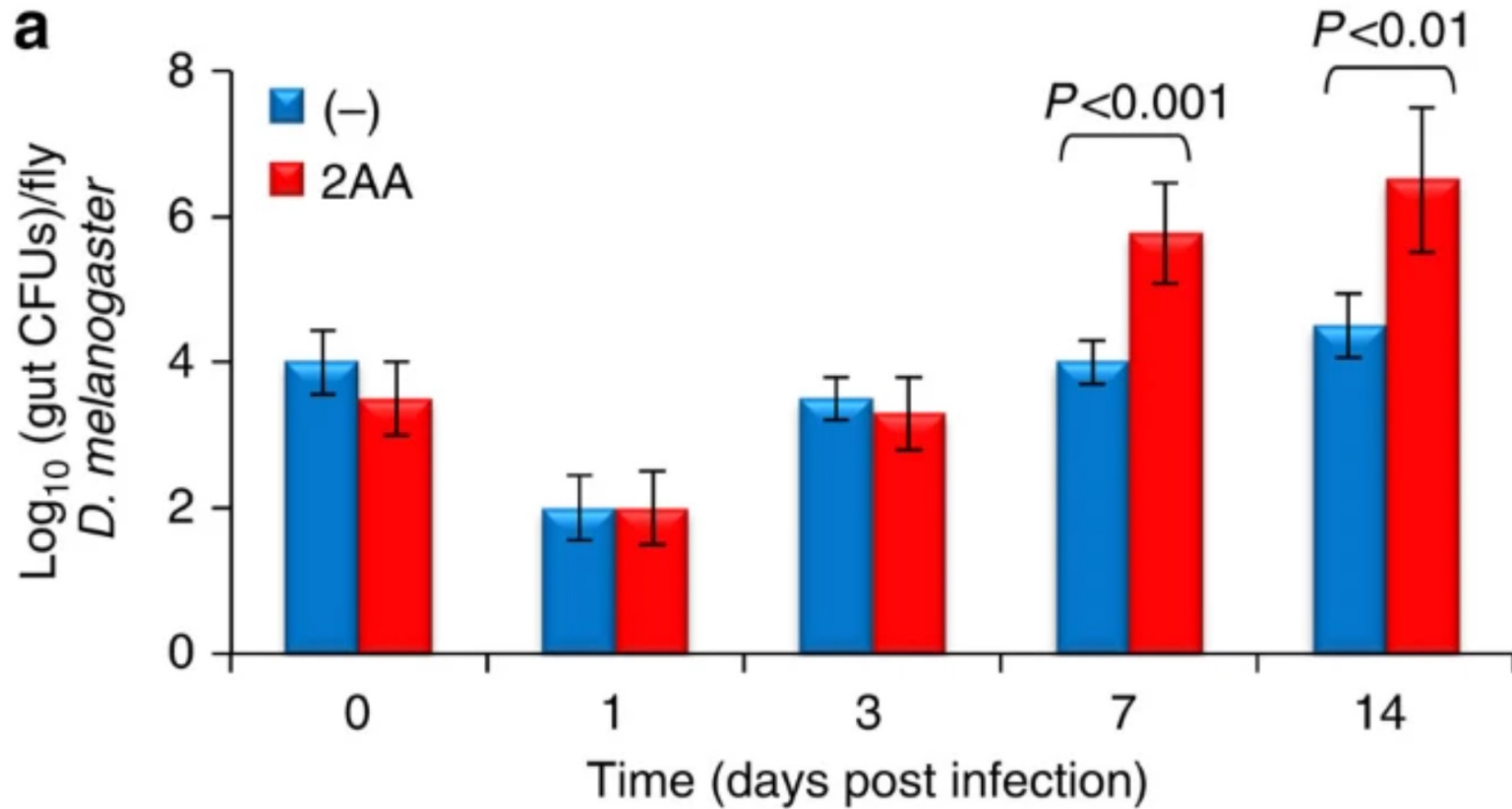
2-AA shifts the cytokine balance toward reduced inflammation, which helps the host survive infection but also creates a permissive environment for bacteria.

The microbial VOC signal also helps to attract the host

2-aminoacetophenone (2AA), secreted by *Pseudomonas aeruginosa* facilitates attraction to food for several fly species including *Musca domestica*, *Ceratitis capitata* and *Drosophila melanogaster*. Constant feeding on 2AA increases the level of long-term colonization of the flies' intestine by *P. aeruginosa*.



2AA lures the flies to the bacterial source and increases the extent of colonization of the fly intestine by the bacteria that produce it, as a means to facilitate bacterial dissemination to new locations.



Bacteria produce and emit highly diverse inorganic and organic volatile compounds

